COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./ DUE DATE
NSF 26-509 07/28/2026

☐ Special Exception to Deadline Date Policy

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NSF PROPOSAL NUMBER
2638522

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OAC - Data Cyberinfrastructure

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NAME OF ORGANIZATION TO WHICH AWARD SHOULD BE MADE
KANSAS STATE UNIVERSITY

AWARDEE ORGANIZATION CODE (IF KNOWN)
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NAME OF PRIMARY PLACE OF PERF
Kansas State University

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TITLE OF PROPOSED PROJECT
Category II: K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science

SHOW LETTER OF INTENT ID IF APPLICABLE

REQUESTED AMOUNT	PROPOSED DURATION (1-60 MONTHS)	REQUESTED STARTING DATE	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE
$ **9,000,000**	36 months	**02/01/2027**	

THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW

☒ TYPE OF PROPOSAL **Research**
☒ COLLABORATIVE STATUS **Non-Collaborative**
☐ BEGINNING INVESTIGATOR
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CERTIFICATION PAGE

Certification for Authorized Organizational Representative (or Equivalent)
By electronically signing and submitting this proposal, the Authorized Organizational Representative (AOR) is: (1) certifying that statements made herein are true and complete to the best of the individual's knowledge; and (2) agreeing to accept the obligation to comply with NSF award terms and conditions if an award is made as a result of this proposal. Further, the proposer is hereby providing certifications regarding conflict of interest, flood hazard insurance, responsible and ethical conduct of research, organizational support, and safe and inclusive working environments for off-campus or off-site research, as set forth in the NSF Proposal & Award Policies & Procedures Guide (PAPPG). Willful provision of false information in this application and its supporting documents or in reports required under an ensuing award is a criminal offense (U.S. Code, Title 18, Section § 1001).

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By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that there is organizational support for the proposal as required by Section 526 of the America COMPETES Reauthorization Act of 2010. This support extends to the portion of the proposal developed to satisfy the Broader Impacts Review Criterion as well as the Intellectual Merit Review Criterion, and any additional review criteria specified in the solicitation. Organizational support will be made available, as described in the proposal, in order to address the broader impacts and intellectual merit activities to be undertaken.

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By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that each individual employed by the organization and identified on the proposal as senior/key personnel has been made aware of the certification requirements identified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 USC 6605(a)(1)).

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AUTHORIZED ORGANIZATIONAL REPRESENTATIVE	SIGNATURE	DATE
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Overview

Computational workflows produce central scientific results, but the systems that run them are usually separate from those that describe, cite, and preserve the scholarly record. Code, data, containers, parameters, hardware assumptions, logs, and provenance remain distributed across registries, schedulers, storage systems, and local documentation. Reuse then depends on reconstruction rather than on a durable and verifiable research object.

Category II: K-FORGE will transition established Kansas State University and regional capabilities into FORGE, a nationally accessible service for publishing, executing, and verifying computational workflows. An InvenioRDM-based registry will create a DataCite workflow record linked to source, contributors, licensing, related research products, build information, and an immutable OCI digest. Beocat will provide managed execution through Slurm/HPC, OpenStack, and Kubernetes; the National Research Platform's Nautilus service will provide an external portability endpoint; and a heterogeneous edge-AI testbed will support selected accelerator-dependent workflows. Each accepted run will produce a verification package containing its execution reference, environment, inputs and outputs, logs, provenance, validation findings, and repository relationships. OPE, K-REx, and New Prairie Press will connect those records to durable public-access and publishing services.

FORGE begins from an operating regional base rather than a new prototype. Beocat serves more than 350 active researchers and operates approximately 10,000 processor cores, 3.3 PB of storage, 170 GPUs, container support, monitoring, and user support. K-REx exposes more than 45,000 public records, K-State Libraries operates a cloud-hosted InvenioRDM test instance, and Sunderland Innovation Lab provides AI and edge-computing prototyping and instruction. Initial pilots will address NABC and AgDEx emergency-data software, executable New Prairie Press research, and a Kansas Water Institute/GeoCat workflow. FORGE will enter full operations by the end of Year 1, expand through the OAK Consortium and Great Plains Network, and then recruit nationally. By the end of Year 3, it targets 250 active users, 25 institutions, 90 registered workflows, and 75 completed verification packages. OEIE will provide independent evaluation, NABC will lead recurring security review, and public reporting will document performance and lessons for other institutions.

Intellectual Merit

FORGE asks what is required to make a computational workflow a citable, executable, and verifiable part of the scholarly record. It will test a standards-based model in which a DataCite DOI identifies the workflow record, an OCI digest identifies the exact executable artifact, and a portable verification package records managed execution. The project will determine which relationships among source, build, version, container, data, hardware, provenance, validation, and publication remain meaningful across registries, repositories, clouds, HPC systems, Kubernetes, and heterogeneous accelerators. It will translate FAIR and FAIR4RS into measurable record requirements and TRUST into measurable service responsibilities. Results will include interoperable metadata and API profiles, operational evidence on portability and reproducibility, and a transferable model for joining research computing with repository and publishing infrastructure.

Broader Impacts

FORGE will make managed workflow verification available to researchers and institutions that cannot operate comparable infrastructure. Public discovery, documentation, APIs, software, metadata profiles, training materials, evaluation products, and implementation guidance will be openly released; primary FORGE-developed software will use the Mozilla Public License 2.0. The service will improve attribution and reuse of research software and give publishers and repositories a practical pathway for executable research. Two graduate researchers in K-State Libraries' Digital Scholarship Training Program will receive mentoring in research software, data stewardship, repositories, cybersecurity, instruction, and responsible research practice. Sunderland and Great Plains Network workshops will extend training to students, researchers, librarians, software engineers, and cyberinfrastructure staff at institutions with different technical capacities. Independent evaluation and public reporting will support other organizations planning trustworthy and sustainable workflow services.

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Category II: K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science

Project Description

1. Vision, Goals, and Driving Requirements

1.1 National Need and Opportunity

Modern research increasingly depends on computational workflows that bring together data, software, and models, increasingly on distributed infrastructure. The systems that execute those workflows, however, are usually separate from the systems that preserve and describe the resulting research record. High-performance computing environments can run demanding analyses but rarely produce durable records of individual runs. Repositories can preserve research products but generally cannot rerun them. Workflow registries improve discovery, but registration alone does not provide managed execution, validation, or long-term public access. Data Curation practices are required to be ported to computational workflows.

The FAIR Guiding Principles may apply to the algorithms, tools, and workflows that produce scientific results as well as to conventional datasets, especially if those workflows are thought of as data themselves. FAIR for Research Software (FAIR4RS) makes that application explicit: research software should have persistent identity and rich metadata, exchange data and metadata through standards and APIs, carry qualified references, licensing, and provenance, and remain both executable and reusable [36, 3]. In practical terms, a reusable workflow needs a persistent identity, machine-readable relationships, clear access conditions, provenance, licensing, and enough technical description to be understood outside the environment in which it was created. The TRUST Principles address the corresponding responsibilities of the service that stewards those records: transparency, responsibility, user focus, sustainability, and appropriate technology [29]. We propose "FORGE (or Federated Open Research Gateway for Executables)" which will use FAIR and FAIR4RS to structure the workflow record and TRUST to structure the service that maintains it. The Kansas State implementation of this stack will be K-FORGE or "Kansas FORGE."

The practical problem is a perceived break in the scientific data lifecycle. A paper may link to source code, a dataset, or a container, while the environment needed to reproduce the result remains scattered across registries, storage systems, scheduler settings, local documentation, and possibly transient logs. The versions, parameters, and dependencies, as well as hardware assumptions, needed to understand the analysis may be incomplete or unavailable. Reuse then depends on reconstruction rather than on a usable scholarly record.

Artificial intelligence and heterogeneous computing make this problem even more visible. AI-enabled workflows may depend on model artifacts, benchmark data, accelerator-specific libraries, and hardware that changes faster than the scholarly record. A container can improve portability without guaranteeing equivalent behavior across processor and accelerator environments. Reproducibility therefore requires more than keeping the code or image. It requires a service that can identify the workflow, execute it under managed conditions, capture what happened, compare the result with an expected outcome, and preserve that evidence with the related research record.

NSF's July 22, 2026 announcement of $83 million in Integrated Data Systems and Services awards places FORGE within a growing national ecosystem of data fabrics, distributed data services, browser-based computing, reusable cloud workflows, and AI-assisted metadata [31]. FORGE will not duplicate those functions. It will provide a verification and scholarly-record layer for workflows and models created or used within them: a persistent record of the workflow, managed re-execution, comparison across computing environments, and durable links to inputs, outputs, provenance, validation findings, and publications.

FORGE will build this service by connecting regional capabilities that already operate at K-State: Beocat research computing, OPE and K-REx repository and public-access services, and Sunderland workflow prototyping and instruction. The award will turn the current handoffs among those services

into a production pathway from workflow registration through managed verification and repository publication.

1.2 Existing Regional Foundation and Category II Transition

FORGE will transition an existing regional service pathway whose constituent capabilities are already in production use. Beocat operates K-State's high-performance computing environment and already supports containerized research workflows through Apptainer. Its public proposal description reports nearly 400 researcher-funded systems, approximately 10,000 processor cores, 3.3 PB of storage, and 170 GPUs; the service website reports more than 350 active researchers and 35 million CPU hours served in 2018 [14, 17]. The operating environment includes Rocky Linux, Slurm scheduling, OpenStack, OpenOnDemand, Globus, software modules, GPU and high-memory resources, connections to ACCESS and the Open Science Grid, and an established Ganglia deployment used to monitor cluster status [16, 18]. Beocat also maintains the K-State namespace, `ksu-nrp-cluster`, on the National Research Platform's Nautilus Kubernetes service, with K-State SSO and policy-governed `kubectl` access [19]. This is an established research-computing service with both local and external execution pathways, not a proposed cluster.

Beocat also has an existing support and operations model. Accounts are normally approved within one business day; users receive maintenance and service announcements through a mailing list; support requests are tracked through K-State's TDX system; and staff provide weekly open support sessions and individual consultation. Beocat reserves scheduled maintenance windows with advance notice and publishes current-status and operating documentation, although it does not publish a consolidated historical uptime percentage [16, 15]. FORGE will therefore use the public service scale, support model, and component inventory as current evidence while establishing auditable availability and incident baselines during transition.

The Open Publishing Exchange (OPE) and K-State Research Exchange (K-REx) create named, discoverable records for software, data, publications, and related scholarly products and support persistent identifiers, metadata, licensing, publishing, and public access. K-REx's public community counts currently sum to 45,016 records: 20,698 electronic theses, dissertations, and reports; 11,261 Libraries Digital Collections records; 9,170 records from academic colleges, departments, and programs; and 3,887 records from additional programs, centers, institutes, and offices. The ETDR total comprises 11,551 records from before 2004 and 9,147 from 2004 onward [24]. K-State Libraries, the lead department for FORGE, also operates a cloud-hosted InvenioRDM test instance. That working environment gives the project an existing base in which to test the DataCite `Workflow` profile, OCI artifact relationships, object-storage patterns, and repository administration before production transition. The Sunderland Foundation Innovation Lab operates an artificial-intelligence studio, NVIDIA Jetson systems, instructional spaces, and an established workshop model that can be extended from introductory edge computing into reproducible AI workflow packaging and validation. Together, these services already support execution, research-object description, public dissemination, prototyping, and training; FORGE will operationally connect those functions into a single verification and scholarly-record pathway.

The existing institutional base also includes personnel who operate these services today. Beocat staff currently comprises of an IT architect, system administrator, and application scientist; Sunderland is comprised of a director and three operational staff; and OPE combines departmental leadership with specialist roles in scholarly communication and copyright, digital scholarship and repository operations, and scholarly publishing and education [20, 26, 25]. The award will add two software engineers and one DevOps specialist in the Beocat reporting line, one instructor in Sunderland, and a project coordinator, administrative assistant, and two graduate researchers in OPE. These eight dedicated hires will expand rather than replace the existing production and user-support base.

The regional demand model is anchored in two established communities rather than an unbounded national-user assumption. The Great Plains Network (GPN) membership consists of 23 full member organizations and four university or network affiliates; its consortium activities explicitly include research computing, cyberinfrastructure, grant collaboration, and training, and it works with national organizations including ACCESS and Internet2 [7]. The OAK Consortium comprises seven founding

institutions in Oklahoma, Arkansas, and Kansas, makes heterogeneous high-performance computing available without charge to researchers across the region, and already convenes recurring conferences and technical workshops [35]. These memberships are an addressable recruitment and dissemination base, although not a claim that all institutions are committed users. FORGE will use GPN as its principal external workshop platform and OAK as an early regional operating cohort before broad national recruitment.

The Category II transition is therefore specific. Current operations can execute containerized workflows, prototype AI and edge-computing activities, and create durable software, data, and publication records, but the handoffs are manual and do not yet produce a unified, digest-bound verification package containing execution evidence, validation results, provenance, and repository relationships. Year 1 will integrate and harden that existing pathway, migrate initial K-State and regional workflows, establish service policies and performance baselines, and enter full operations. Years 2 and 3 will scale the accepted service through OAK and GPN institutions and then nationally, while improving interoperability, reliability, user support, and sustainability. FORGE is therefore the production transition of an existing regional execution-to-publication pathway into a nationally accessible verification service.

1.3 Vision

FORGE treats the computational workflow as a scholarly object in its own right. The durable center of the service will be a DataCite metadata record with `resourceTypeGeneral=Workflow`. That record can carry the DOI, creators and contributors, title, publisher, publication year, subjects, descriptions, version, rights, funding, and qualified relationships to the data, software, models, publications, and other objects needed to understand the workflow [6]. The exact executable image will be identified separately by its immutable OCI digest. In this model, the DataCite record identifies and cites the workflow; the digest identifies the precise container artifact; and a verification package records what happened when that version was executed.

FORGE is not designed as a new workflow language, a replacement for GitHub or GitLab registries, a general repository, or another computing cluster. Its contribution is the connection among a citable workflow record with permanent identifiers (PIDs), an exact executable artifact, a managed verification run, and a durable, citable repository record with permanent identifiers (PIDs) and provanance. In FORGE operations FAIR principles provide the object-level requirements for discovery and reuse; TRUST principles provide the operating requirements for stewardship, continuity, and accountability. FORGE will use established interfaces wherever possible, including an OCI-compatible registry and standard CI/CD workflows, so that it can interoperate with rather than replace existing research infrastructure.

1.4 Project Goals

FORGE will pursue four integrated goals.

- Goal 1: Establish a DataCite-centered workflow registry with OCI-compatible distribution. Each registered workflow version will have a citable scholarly record and qualified links to its container image, source, data, models, documentation, publications, and verification evidence. The registry will support versioning, immutable digest identification, discovery, standards-based push/pull and import/export interfaces, and automated publication from GitHub Actions, GitLab CI/CD, and comparable systems.
- Goal 2: Operate managed verification pathways for reproducible and AI-enabled workflows. FORGE will connect registered or externally supplied workflows to managed Kubernetes, Slurm/HPC, virtualized service, and selected heterogeneous-computing pathways. The service will provide staging, scheduling, policy and security review, monitoring, logging, telemetry, output capture, reliability testing, validation, and user support.
- Goal 3: Connect verification evidence to the scholarly record. A completed run should not end as a directory of cluster files and logs. FORGE will create or link durable records for the workflow, its environment, inputs, outputs, provenance, validation findings, documentation, contributors, license, and associated publications.

- Goal 4: Transition the regional service base to trustworthy national operations and produce a transferable verification-service model. FORGE will onboard transdisciplinary users and communities, operate the integrated service by the end of the first award year, demonstrate interoperability with complementary data and workflow platforms, evaluate user experience and service performance, publish implementation and operational guidance, and document the governance, staffing, security, sustainability, and partnership models needed for adoption by other institutions.

1.5 Driving Requirements

1. National and transdisciplinary accessibility. The service must support users and use cases from multiple scientific and engineering domains and must not depend on a single discipline-specific data model, workflow engine, data fabric, or computing environment.
2. Operational readiness. The initial integrated service must enter full operations by the end of Year 1. Years 2 and 3 will emphasize continuous operations, national onboarding, performance improvement, expanded interoperability, and controlled innovation during operations.
3. End-to-end verification lifecycle. The service must connect onboarding, packaging, registration, policy review, staging, execution, capture, validation, repository integration, dissemination, and reuse.
4. Interoperability and portfolio complementarity. FORGE must use standards-based interfaces and interoperate with existing repositories, workflow registries, data fabrics, identity systems, research computing resources, and national cyberinfrastructure services. The project must demonstrate its unique role without duplicating newly funded IDSS systems or other established platforms.
5. Developer-native registry and automation integration. FORGE must support standard OCI clients, immutable digest references, scoped authentication, and documented continuous integration and deployment patterns. Researchers and software teams must be able to publish or register container images from GitHub Actions, GitLab CI/CD, and comparable systems without adopting a FORGE-specific build tool.
6. FAIR workflow records and TRUST-aligned stewardship. The workflow record must be persistently identifiable, sufficiently described, retrievable through standard protocols, and connected to related objects through qualified relationships. Reuse requires a clear license, version, provenance, documentation, and validation evidence. The service must publish its scope and policies, assign stewardship and security responsibilities, incorporate user feedback, and plan for continuity. Each verification package must show what was executed, with which inputs and environment, under what conditions, and with what result.
7. Security, trustworthiness, and policy awareness. The service must support identity, authorization, auditability, software and workflow review, data-use constraints, node and institutional policies, incident response, and documented risk management.
8. Usability, supportability, and sustainability. The service must be usable by researchers who are not workflow-infrastructure specialists, supported through documented onboarding and service procedures, measurable against operational targets, and sustainable through realistic staffing, governance, partnership, and cost models.

1.6 Initial Pilot Workflows

FORGE will begin with three named pilot pathways spanning policy-aware agricultural software, executable publication, and geospatial water research. Each will produce a DataCite workflow record, immutable artifact reference, managed verification run, and repository-linked verification package.

NABC/Ag-DEx emergency-data workflow publication pilot. Adrian Self will lead the domain and security relationship through NABC. FORGE will package a bounded software component emerging from Ag-DEx planning or related NABC emergency-data-management work, such as validation, compatibility mapping, controlled exchange, scenario analysis, or generation of non-sensitive decision products. Ag-DEx addresses research-data management and governance; this pilot tests the distinct FORGE task of turning resulting software into a registered, executable, and verifiable research object under realistic agricultural-security controls [13, 27, 28].

New Prairie Press executable-publication pilot. David Molik and Gwendolyn Sibley will lead the first implementation using the next suitable software- or methods-bearing submission in *Cyberbiosecurity* or the *Journal of Research in Rural Education* (JRRE), then convert the tested pathway into a supported NPP service [32, 33, 34]. FORGE will connect the article DOI to a DataCite Workflow DOI, immutable OCI artifact, source and build records, and Beocat verification package, testing editorial intake, rights review, author documentation, repository deposit, and post-publication rerunning.

Kansas Water Institute/GeoCat workflow-portability pilot. Susan Metzger will serve as the KWI intermediary, connecting FORGE with GeoCat and the appropriate scientific and technical leads. GeoCat is K-State's cloud-based ArcGIS Enterprise and research-data-management environment, supported by KWI and AWS with GISSAL participation [21, 22, 23]. A selected geospatial or water workflow will be registered and compared across GeoCat/AWS and Beocat, with optional Jetson testing where useful, creating a citable record for cloud geoprocessing, modeling, remote sensing, GeoAI, or derived decision products.

Under the supervision of Molik, the two OPE graduate researchers will use these pilot workflows as their primary technical practicum. Specific software components and participating articles will be finalized in Q1 using readiness, data availability, rights, scientific value, and ability to complete end-to-end verification as selection criteria. The NABC pilot does not depend on Ag-DEx receiving an award; if needed, NABC will designate an equivalent bounded emergency-data software workflow.

2. Project Definition and Specification

2.1 Service Definition

FORGE is a verification and scholarly-record service for computational workflows. It combines a DataCite-centered workflow registry, OCI-compatible container distribution, managed execution on production research computing, and repository publication of verification evidence. Operations, training, security review, and independent evaluation are part of the service rather than separate outreach activities.

The design makes a deliberate distinction between the record and the service. DataCite metadata, repository records, and portable research-object packages describe the workflow and its relationships. TRUST-aligned operating policies describe who is responsible for the record, what the service retains, how users gain access, how incidents are handled, and how the service will continue beyond the award.

The service will publish its policies, onboarding and support procedures, performance targets, security controls, governance, and sustainability plan. Project software will be released under an identified open-source license. FORGE will expose an OCI Distribution Specification-compatible endpoint and documented APIs for image publication, workflow registration, verification requests, status and log retrieval, and export of repository-ready records. The documentation will include tested GitHub Actions and GitLab CI/CD examples that can be copied into an existing repository with minimal modification.

2.2 Intended Users

FORGE will serve researchers who publish, rerun, validate, adapt, or benchmark computational workflows; software developers and research software engineers who build executable environments through GitHub Actions, GitLab CI/CD, or comparable automation; and data stewards, repository professionals, publishers, and cyberinfrastructure staff who connect computation to durable scholarly records. It will also support users of national data fabrics, workflow platforms, and computing gateways that need an independent verification and publication pathway, as well as educators, students, regional institutions, and research communities that cannot independently operate comparable infrastructure. During the award, eligibility will be open to the U.S. research and education community, subject to published security, data-policy, technical-feasibility, and resource-allocation rules.

2.3 National Operations, Access, and Community Onboarding

FORGE will expand through a staged operating model: K-State workflows will supply the earliest production cases; OAK and GPN institutions will form the first external cohort and primary workshop audience; and national recruitment will broaden after the Year 1 acceptance review. This sequence manages operational risk and support load but does not restrict eligibility by institutional affiliation. GPN's existing research-computing, cyberinfrastructure, security, and training communities provide a practical channel for testing documentation, onboarding, and service policies across institutions with different capacities [7]. OAK provides a complementary early cohort with an established three-state computing and training mission [35].

Access will be broad, open, and transparent. Public documentation, metadata discovery, registry search, published verification records, training materials, and standard APIs will be available without an allocation. Routine verification requests that fit published limits will enter a fair-share queue under standard quotas and time windows. Requests requiring substantial staff engineering, scarce accelerators, unusually large data movement, controlled data, or sustained compute will undergo a documented technical and merit review based on scientific or educational purpose, reproducibility value, readiness, security and policy fit, feasibility, resource efficiency, and equitable access. Decisions, expected turnaround, limits, and appeal or resubmission procedures will be published. No user fee is planned during the award; the sustainability plan will distinguish the institutionally supported K-State core, shared OAK/GPN activities, and national services that may later use renewed federal support, consortium agreements, partner contributions, or transparent cost recovery for unusually intensive workloads.

The first cohort will receive early access during Year 1 through three stages: closed integration testing with K-State workflows, invited OAK/GPN testing, and an external beta available to qualified U.S. users. Project staff and OEIE will collect workflow success, support effort, metadata completeness, user experience, security findings, and resource consumption at each stage. The leadership team will approve progression only after corrective actions are closed; the final Year 1 Operations Plan, quotas, service targets, and documentation will be revised from this evidence before full acceptance.

Developer-native onboarding will use familiar container registry and CI/CD patterns. A project may push an image or other OCI artifact directly to a FORGE namespace, or register an immutable digest reference to an image hosted in GitHub Container Registry, the GitLab Container Registry, or another compatible OCI registry. Lightweight connector profiles will allow data fabrics, gateways, workflow workspaces, and repositories to submit or export a research object for verification and receive a standards-based record of execution, provenance, validation, and repository relationships without adopting FORGE as their primary interface.

Sustainability will be treated as an operating requirement from project initiation rather than as an end-of-award transition activity. FORGE will maintain an annually revised sustainability plan that identifies the personnel, infrastructure, storage, security, training, and user-support costs associated with each service component; distinguishes essential public infrastructure from resource-intensive optional services; and assigns responsibility for maintaining each component after the award. The plan will use evidence from workflow demand, staff effort, compute and storage consumption, partner participation, and user outcomes to determine which functions should remain part of K-State's institutional core, which can be shared through OAK, GPN, or other consortium arrangements, and which require renewed federal investment, external sponsorship, partner contributions, or transparent cost recovery. Open standards, portable workflow records, documented APIs, modular software, and exportable metadata will reduce dependence on any single platform or funding source and will allow participating institutions to adopt individual FORGE components without assuming responsibility for the entire service. Beginning in Year 2, the leadership team and advisory committee will review the sustainability plan annually and publish recommended service tiers, partnership responsibilities, transition thresholds, and post-award commitments as part of the Operations Plan.

2.4 Reference Scientific Data Lifecycle

1. researcher, community, or external-platform onboarding;

2. workflow and data assessment;
3. packaging or import of software, workflow definitions, models, metadata, licenses, and documentation, including direct CI/CD publication of container images or OCI artifacts and registration of immutable external-registry references;
4. registration of objects, versions, image digests, tags, source repositories, commits, build workflows, attestations, dependencies, identifiers, and relationships;
5. policy, security, and execution review;
6. staging of inputs through repository, object-storage, transfer, data-fabric, or data-visitation mechanisms;
7. managed execution through Kubernetes, Slurm/HPC, virtualized services, or controlled heterogeneous pathways;
8. capture of logs, telemetry, environment fingerprints, resource use, errors, outputs, and provenance;
9. validation against expected results, previous runs, or cross-environment comparisons;
10. repository deposit or linking of workflows, inputs, outputs, provenance, documentation, validation records, and publications;
11. export of a machine-readable verification package to the originating platform or downstream service; and
12. evaluation, support, dissemination, and reuse.

2.5 Architecture

FORGE will transition K-State Libraries' existing cloud-hosted InvenioRDM test environment into a maintained production deployment, using a software-fork where needed, that serves as the metadata, DOI, landing-page, versioning, access-control, and repository layer for workflow records. InvenioRDM supports customizable resource-type vocabularies and DataCite DOI registration; FORGE will define a `Workflow` resource type mapped to DataCite Metadata Schema 4.7 rather than build a repository framework from scratch [9, 8]. The record will carry the DOI, creators and contributors, title, publisher, publication year, subjects, dates, version, rights, descriptions, funding, and qualified relationships to data, software, models, documentation, prior versions, publications, and verification packages [6]. More detailed run-level relationships will be carried in RO-Crate, Workflow Run RO-Crate, provenance records, and repository metadata.

The InvenioRDM deployment will add an OCI external-artifact adapter. To the user, an immutable container image will appear as a resolvable artifact associated with the workflow record; the container manifest and layers will remain in the OCI registry rather than being copied into ordinary repository file storage. The adapter will validate and record the registry locator, digest, media type, size, tag aliases, signatures, software bills of materials, and attestations, and will expose controlled resolution through the record and API. InvenioRDM's extensible uploader, external-file, and storage-location mechanisms provide the implementation points for this work [11, 10, 12]. Human-readable documentation and bounded verification packages may remain in InvenioRDM's S3-compatible storage. If the full artifact adapter is not ready at Year 1 acceptance, the service will use a qualified external identifier and digest-resolution API without changing the DOI, metadata, or execution model.

The DataCite DOI and the OCI digest serve different purposes. The DOI identifies the citable workflow record and resolves to its InvenioRDM landing page. The OCI digest identifies the exact executable artifact. K-REx or another recognized repository will preserve significant software releases, schemas, metadata exports, and verification packages as described in the Data Management and Sharing Plan. This separation permits the scholarly record to remain stable even when a mutable tag changes or an operational copy moves between registries.

This approach implements FAIR in practical terms. The DOI and indexed metadata make the workflow findable. Resolvable identifiers and standard protocols make the record accessible under stated conditions. DataCite relationships, OCI interfaces, and portable RO-Crate packages make it interoperable. Version, rights, provenance, documentation, digest references, and verification evidence make it reusable [36]. TRUST applies to the service that maintains those records. FORGE will publish its scope, retention boundaries, service targets, known limitations, and evaluation

findings; assign responsibility for curation, security, integrity, and incident response; incorporate user feedback; and maintain documented continuity and sustainability plans [29].

The container registry remains an operational distribution service, not the scholarly record itself. FORGE will expose an OCI Distribution Specification-compatible endpoint for approved namespaces and accept standard clients such as Docker, Podman, Buildah, and ORAS. Images and other OCI artifacts will be resolved by immutable digest; tags will remain convenient, mutable aliases. The registry will retain or validate OCI annotations and, where supported, connect signatures, software bills of materials, and build-provenance attestations through OCI referrers or compatible mechanisms.

FORGE will support two publication paths. A GitHub Actions or GitLab CI/CD workflow may build and push an image directly to a FORGE namespace and then register the resulting digest. A project may instead keep the image in GitHub Container Registry, GitLab Container Registry, or another OCI-compatible service and submit the immutable digest reference to FORGE. In either case, the registration request will create or update the DataCite workflow record, attach the source and build metadata, and request verification. Maintained example repositories will document least-privilege authentication, digest capture, metadata submission, verification status, and returned DOI and repository links.

Registry operations will include namespace and role management, scoped service credentials, audit logging, retention and deletion rules, optional immutability or promotion policies, backup and recovery, storage accounting, security and vulnerability-review integration, and incident response. OCI conformance testing, standard-client interoperability, and successful GitHub Actions and GitLab CI/CD publication tests will be included in Year 1 operational-readiness criteria.

The execution layer will use a hybrid model. Beocat's Slurm/HPC environment will support batch and research-computing workloads; local Kubernetes and OpenStack resources will support container-native execution and persistent services; and the existing `ksu-nrp-cluster` namespace on NRP Nautilus will provide an initial external Kubernetes endpoint for portability and federation testing [19]. Nautilus will be integrated through a policy-aware connector and used for approved workloads, but it will not be a single point of failure for core FORGE operations. A controlled heterogeneous testbed will support selected edge-AI compatibility and reproducibility studies. Workflow engines and orchestration tools will be selected based on pilot requirements, interoperability, maintainability, security, and community adoption.

The interoperability layer will keep FORGE's verification functions separate from user-facing portals and upstream data services. Its APIs will support OCI distribution and digest resolution, CI/CD metadata submission, import and export of workflow and run packages, data transfer or visitation, submission to approved execution backends, return of status and validation results, and deposit or linkage to repository records. This boundary lets FORGE work with developer-native registries, browser environments, workflow platforms, distributed data services, metadata systems, and Category I data fabrics without reproducing their primary functions.

The storage and repository layer will separate container distribution from repository stewardship. The OCI registry will hold manifests and layers; InvenioRDM will hold workflow metadata, DOI landing records, documentation, bounded verification packages, and external-artifact relationships; project storage will hold active inputs, outputs, logs, provenance, attestations, benchmarks, and staging copies. DataCite and repository records will remain available under published policy even when an operational image or staging copy is no longer retained. FORGE will not promise permanent hosting for all research data or every development image. Long-term preservation will use the FORGE InvenioRDM service, K-REx, partner repositories, defined retention policies, and post-award service agreements.

The federation and policy layer will support institutional identity, delegated authority, auditable registration and execution events, local data-use policy, and secure exchange among participating services. Production operations will begin at K-State, but the APIs, verification profiles, and node-onboarding documentation will permit additional institutional execution sites. GPN will provide the first regional test of that model through advisory review, workshops, and member engagement. Signed metadata, transparency logs, verifiable credentials, graph databases, and other

trust mechanisms remain implementation choices to be evaluated; none is a prerequisite for the core service.

2.6 Service Continuity, High Availability, and Cost Alignment

FORGE will use a layered continuity architecture in which each major infrastructure investment has a distinct operational purpose. This architecture implements the TRUST expectations of sustainability and appropriate technology while preserving the solicitation's boundary between active operational storage and permanent long-term hosting. The project will not operate two complete production environments or use cloud services to duplicate the Beocat execution platform.

Component	Primary role	Boundary and non-duplication
Beocat execution environment	Primary managed execution for Kubernetes, Slurm/HPC, workflow staging, logging, telemetry, output capture, and validation.	Cloud continuity will not reproduce the full Beocat compute environment or provide a second general-purpose HPC service.
OpenStack service nodes	Primary hosting for InvenioRDM, the OCI registry control plane, APIs, identity, metadata, databases, monitoring, and related persistent services.	These nodes separate durable control and application functions from transient research workloads.
Repository and workflow storage	Active storage for OCI registry manifests and blobs, workflow inputs, reference data, logs, provenance, attestations, outputs, benchmarks, validation packages, and repository staging.	This storage supports operations and ingest; it is not a promise of indefinite preservation for every generated object.
Local power, network, and HA components	Reduce project-specific single points of failure through redundant power paths, rack-level distribution and monitoring, component redundancy, and a second network path.	Funds will support only FORGE-attributable equipment, installation, testing, and commissioning, not general building renovation or unrelated deferred maintenance.
AWS GovCloud warm continuity environment	Maintain selected backups, replicas, monitoring, recovery data, and limited failover for critical InvenioRDM, registry, and application functions under K-State's institutional AWS agreement.	GovCloud is a recovery and compliance-readiness layer, not a duplicate production storage system or execution service.
K-REx and preservation partners	Provide durable public-access records, persistent identifiers, metadata, and preservation handoffs for selected research objects and run evidence.	Repository preservation is distinct from active execution and staging storage.

During Year 1, facilities and data-center personnel will confirm the power load, second-UPS and distribution design, network path, rack integration, monitoring, cooling compatibility, and commissioning requirements for the equipment acquired by the project. Only allowable, project-attributable components and associated installation will be charged to the award. FORGE will not fund general renovation, expansion of campus utility capacity, or correction of unrelated facilities deficiencies. The planned warm continuity environment will use AWS GovCloud (US) through K-State's institutional

cloud agreement. It will be sized from recovery objectives, monitored for cost, and kept as a limited continuity layer. GovCloud's isolated U.S. regions and compliance programs will support a security-control mapping and independent readiness program; FORGE will pursue a certifiable operating posture and appropriate external attestations without treating use of GovCloud as certification of the service itself [1, 2].

2.7 Portfolio Alignment, Complementarity, and Unique Contribution

The newly announced IDSS portfolio makes non-duplication a central design requirement. FORGE will treat the funded systems as complementary infrastructure and potential interoperability targets. It will not claim novelty for general data discovery, national data fabrics, browser-based access to computing, reusable cloud workflows, or AI-assisted metadata creation. Its unique contribution is the operational connection among persistent registration, managed verification execution, run-level evidence, validation, repository and public-access integration, heterogeneous portability, and independent service evaluation [31].

IDSS service	Announced emphasis	FORGE boundary and interoperability
FabAID	National data fabric connecting repositories, computing, and large scientific datasets.	Use discovery, transfer, or visitation interfaces where appropriate; return persistent verification records. FORGE will not build a competing national data fabric.
National Data Platform	Distributed data, computing, and AI resources for AI-ready workflows.	Accept workflow or model packages and return digest-bound execution and repository evidence rather than duplicate a national AI ecosystem.
iDLab	Browser-based access to NSF HPC and cloud resources.	Expose verification as a backend or export target; do not build a general browser portal.
BRIDGE	Reusable cloud workflows, datasets, and machine-learning models.	Import and export open workflow packages; verify selected workflows across HPC, Kubernetes, and heterogeneous environments.
National Science Data Fabric	National discovery and access to distributed scientific data.	Use data delivery or visitation mechanisms as inputs and record what was executed against which data and policy.
MESA	AI-assisted metadata organization for multidisciplinary data.	Exchange metadata with provenance controls and preserve run-specific identity, validation, and scholarly relationships.

FORGE will also complement WorkflowHub, Dockstore, Whole Tale, Binder/JupyterHub, Galaxy, Renku, Code Ocean, InvenioRDM, Dataverse, Globus, ACCESS, PATh/OSG, and related services. Developer-native systems including GitHub Container Registry and GitHub Actions, GitLab Container Registry and GitLab CI/CD, and other OCI-compatible services will remain upstream build and distribution environments; FORGE will add digest-bound scientific metadata, policy-aware managed verification, run-level evidence, validation, and repository linkage. This standards-based boundary allows FORGE to remain useful even when another platform's API or schedule changes and creates a route for products generated elsewhere to become citable and verifiable.

3. Concept of Operations and Operations Plan

3.1 Modes of Operation and Access

FORGE will operate as a centrally managed national service. Researchers may create or discover a DataCite workflow record, publish an OCI artifact or reference an external digest, request managed verification, and receive a repository-linked verification package. External systems will use the same functions through documented APIs. Discovery and documentation will be open; execution will require authentication, accepted terms, and a standard or reviewed allocation as described in Section 2.3.

The operating model will scale in phases from K-State to OAK/GPN and then nationally. This is a support and acceptance sequence rather than a closed membership model. Routine requests will

use published quotas and fair-share scheduling. Resource-intensive or policy-sensitive requests will be reviewed transparently, with decisions based on merit, feasibility, reproducibility value, security and data-use constraints, and equitable access. Service status, aggregate usage, performance against targets, known limitations, and scheduled maintenance will be published on a public dashboard without exposing protected user or research information.

3.2 User Journey

A user will enter through documented onboarding, an approved external-system connector, or a CI/CD publication workflow. Staff will assess intended use, data sensitivity, software and licensing status, expected outputs, resource requirements, registry location, build provenance, and readiness for managed verification. Workflows requiring substantial translation may first use Sunderland prototyping and training before Beocat-facing execution.

Staff will create or update the InvenioRDM workflow record, register or reserve its DataCite DOI, and connect it to the immutable image digest, source repository and commit, build workflow, dependencies, expected inputs and outputs, license, provenance requirements, and related data and publications. Policy and security review will determine whether data may be transferred, mounted, staged, or visited in place. Managed execution will capture the environment, resource use, logs, errors, outputs, and run-level provenance. Validation will compare those results with an expected outcome, prior run, or alternate environment and produce a durable verification package linked back to the workflow DOI.

3.3 Early Operations, Acceptance, and Full Operations

Year 1 will progress through quarterly readiness gates. Q1 will establish baselines, hire staff, finalize architecture and procurement, and select K-State pilot workflows. Q2 will deploy the InvenioRDM fork, OCI registry, artifact adapter, and core APIs; connect initial Beocat and K-REx pathways; and begin closed testing. Q3 will open an invited OAK/GPN beta, complete security and AWS GovCloud recovery exercises, and test GitHub Actions, GitLab CI/CD, OCI clients, the Nautilus execution connector, and at least one additional external import/export profile. Q4 will resolve findings, conduct an independent operational-readiness review, publish final service policies and targets, and enter full operations by the end of Month 12.

Acceptance will require successful end-to-end execution and repository linkage for the initial pilots; OCI distribution conformance and standard-client tests; authenticated direct-push and external-digest workflows; backup and recovery exercises; closure of critical security findings; published access, retention, support, incident, and maintenance policies; complete user and operator documentation; and achievement of the initial Year 1 targets in Section 4. Early-user evidence will be summarized by OEIE and used to adjust quotas, support procedures, documentation, staffing priorities, and the final Operations Plan. Years 2 and 3 will maintain continuous service while expanding users, integrations, training, reliability, and sustainability.

3.4 Personnel, Reliability, and User Support

The operating base combines existing service personnel with eight dedicated hires. Beocat's current IT architect, system administrator, and application scientist will be augmented by two software engineers and one DevOps specialist. Sunderland's director and three operational staff will be augmented by one instructor. OPE's department leadership and specialist librarians will be augmented by a project coordinator, administrative assistant, and two graduate researchers. The project coordinator will run the intake and support process; Beocat personnel will provide infrastructure and execution support; OPE personnel will support metadata, rights, persistent identifiers, repository handoffs, and documentation; Sunderland personnel will provide prototyping and instruction; NABC will lead recurring security review; and OEIE will independently assess readiness, use, and outcomes [20, 26, 25].

FORGE will extend rather than replace Beocat's current user-support and monitoring model. Public documentation, training videos, a tracked TDX queue, weekly office hours, account provisioning, maintenance announcements, individual consultation, and Ganglia cluster-status monitoring already

support Beocat users [16]. FORGE will use Ganglia metrics as one input to a broader observability layer that also covers InvenioRDM, registry, API, job, storage, connector, and verification-package health. It will add a coordinated intake path, workflow-specific documentation, repository and rights support, and escalation among OPE, Sunderland, Beocat, and NABC. Current maintenance practice permits announced windows as often as every other week; FORGE availability targets will exclude approved maintenance and will be measured independently rather than inferred from undocumented historical uptime [15].

Support will use a documented escalation model. Public documentation, examples, status information, and office hours will address common needs. The project coordinator and administrative assistant will triage requests and track response times. Graduate researchers and the Sunderland instructor will assist with packaging, examples, and workshops under supervision. Complex execution, software, security, repository, or policy questions will be escalated to the responsible technical lead. Service procedures will cover monitoring, incident response, backup and recovery, change management, workflow approval, software maintenance, registry credentials and namespaces, retention and deletion, metadata integrity, connector maintenance, and repository handoff.

An AWS GovCloud warm environment, operated through K-State's institutional cloud agreement, will support selected backups, monitoring, recovery data, and limited failover for critical InvenioRDM, registry, and application functions. Primary workflow execution will remain on Beocat; Nautilus will provide an external Kubernetes endpoint for approved portability tests; active workflow data will remain on project storage; and durable public-access records will be managed through InvenioRDM, K-REx, and preservation partners. Recovery exercises, incident metrics, and availability will be reported against the targets in Section 4.

3.5 Technology Refresh, Innovation, and Training

FORGE will use a development, staging, and production release process with automated tests, infrastructure as code, versioned configurations, scheduled maintenance windows, rollback procedures, and documented API deprecation periods. The leadership team will review dependencies, container runtimes, workflow connectors, registry software, security patches, accelerator support, and hardware health quarterly. Urgent security fixes may use an expedited change process; other major upgrades will be tested with representative workflows before production release. A bounded innovation track will evaluate new accelerators, metadata approaches, identity mechanisms, and external connectors without placing accepted production service at risk.

Sunderland will use its AI studio and NVIDIA Jetson systems to teach container construction, edge-AI packaging, CI/CD publication, portability testing, and reproducibility. GPN will be the principal external workshop platform, with OAK providing an early regional audience. Maintained tutorials will cover GitHub Actions and GitLab CI/CD, immutable digests, build metadata, verification requests, FAIR research-object packages, and interpretation of validation and repository records. Training will emphasize movement among data, computing, workflow, and repository systems while preserving identity, policy, provenance, and citation relationships.

4. Performance Objectives and Measures

FORGE will use technical monitoring, service analytics, user feedback, project records, quarterly advisory input, and OEIE-led evaluation to assess implementation and outcomes. The targets below are operating commitments derived from the high-touch nature of workflow verification, the proposed staffing model, and the reachable OAK/GPN community; they do not assume that all GPN or OAK members will become users. The current public scale of Beocat and K-REx is documented in Section 1.2. During mobilization, FORGE will establish a common operational baseline across execution, registry, repository, training, support, and evaluation functions; any material change to an award target will require evidence, written justification, Advisory Committee review, and NSF concurrence when required.

4.1 Evaluation Logic, Independence, and Public Reporting

OEIE will combine independently collected evidence with technical data produced by the service. Evidence will include readiness reviews, availability and incident records, workflow and metadata quality, onboarding and support records, user surveys and interviews, training feedback, quarterly advisory recommendations, pilot case studies, interoperability tests, and cost and staffing data. OEIE will validate measures, assess whether readiness and adoption claims are supported, and prepare formative, annual, summative, and public reports. FORGE will publish a quarterly aggregate dashboard covering users, institutions, disciplines, workflows, verification outcomes, availability, support responsiveness, training, and major incidents, while protecting sensitive information.

4.2 Initial Service and Adoption Targets

Measure	End of Year 1	End of Year 2	End of Year 3
Service state	Full operations by Month 12; all acceptance gates met	Continuous operations and annual review	Continuous operations and sustainability transition
Availability after acceptance	≥99.0% excluding announced maintenance	≥99.5%	≥99.5%
Platform-caused completion rate for accepted standard workflows	≥85%	≥90%	≥92%
Routine support first response	90% within two business days	90% within one business day	95% within one business day
Registry/control-plane recovery objectives	RTO ≤8 hours; RPO ≤24 hours	RTO ≤6 hours; RPO ≤12 hours	RTO ≤4 hours; RPO ≤12 hours
Active users	40	120	250
Participating institutions	6, including external OAK/GPN users	15	25, including at least five outside OAK/GPN
Represented disciplines	4	6	8
Registered executable workflows	15	45	90
Completed verification packages	10	35	75
External workshops / participants	2 / 50	4 / 120	4 / 180
Advisory meetings	4	4	4
User satisfaction	≥80% satisfied or very satisfied	≥85%	≥85%

Active users are authenticated individuals who complete onboarding or use a registry, verification, repository, or training function during the reporting year. Institution counts are unduplicated organizations. Workshop participation is reported separately from active service use. The Year 3 institutional target represents less than the current GPN membership base plus national recruitment and is therefore ambitious but plausible for an eight-position expansion supported by existing unit staff [7, 35].

4.3 Reproducibility, FAIRness, and Scholarly Integration

A successful verification package must contain an immutable execution reference; source and build relationships; identifiers for the workflow and contributors; machine-readable metadata; documented inputs, outputs, environment, and access conditions; a license or rights statement; run-level provenance; validation status and deviations; and a durable repository record or resolvable preservation handoff. At least 90% of accepted packages will meet all required elements in Year 1, 95% in Year 2, and 97% in Year 3. At least 80%, 85%, and 90% of workflows accepted for repeat verification will rerun successfully or produce a documented, scientifically interpretable deviation in Years 1, 2, and 3, respectively. OEIE will report FAIR maturity by element rather than claim binary certification.

4.4 Interoperability and Portfolio Complementarity

By the end of Year 1, FORGE will demonstrate an operational InvenioRDM Workflow profile; OCI distribution conformance; tested direct publication from GitHub Actions and GitLab CI/CD; immutable registration of images in GitHub and GitLab registries; K-REx deposit or linking; execution through the Beocat Nautilus connector; and at least one additional external import/export profile using an open research-object, workflow, repository, transfer, or data-access interface. The cumulative targets are four maintained external integration profiles in Year 1, six in Year 2, and eight in Year 3. Candidate profiles include RO-Crate and Workflow Run RO-Crate, WorkflowHub or Dockstore exchange, Globus-assisted movement, InvenioRDM-compatible repository exchange, and interfaces published by IDSS portfolio services. A portfolio-service demonstration will be pursued when an interface and permission are available; an equivalent open external service will be used if another award's schedule prevents dependency-free testing.

4.5 Governance, Trustworthy Operations, and Sustainability

The six-member Advisory Committee will meet quarterly, with the PI and co-PIs participating in all four meetings. One quarterly meeting will include the project's annual PI/co-PI strategic and performance review and preparation for or follow-up from the required NSF PI meeting or designated cyberinfrastructure conference. Measures will include attendance, recommendations, action closure, response to user and OEIE findings, risk closure, policy publication, Ganglia and service-dashboard coverage, AWS GovCloud recovery exercises, security and certification-readiness assessments, continuity and succession planning, cost and staffing models, external integration agreements, and institutional sustainability commitments. TRUST evidence will be reported for transparency, responsibility, user focus, sustainability, and fit-for-purpose technology rather than as a binary certification.

4.6 Workforce and Broader Outcomes

FORGE will track students and professionals trained, experiential learning, documentation and software contributions, workflow packages, workshop participation, and subsequent use of the skills or service. Graduate-researcher measures will include Individual Development Plans, quarterly reviews, monthly Digital Scholarship Seminar participation, technical and scholarly products, supervised teaching progression, self-assessment, and confidential feedback. OEIE's public report will document the operating model, pilot workflows, user findings, readiness, interoperability, limitations, and lessons for institutional and national research infrastructure communities.

5. Project Management

5.1 Project Leadership Team and Their Experience

David Molik, Ph.D., Principal Investigator and OPE department head, will serve as the project's executive authority and will hold final responsibility for scope, budget, schedule, sponsor communication, integration, governance, dissemination, and sustainability. Nicholas Eggleston, Co-PI and Beocat IT architect, will lead production research computing, infrastructure, execution, monitoring, reliability, and operational readiness and will report directly to the PI for award scope and performance. Jeffrey Sheldon, Co-PI and Sunderland director, will lead prototyping, instruction, student participation, documentation, workshops, and heterogeneous-computing activities and will report directly to the PI for award scope and performance. Gwendolyn Sibley, named key personnel, will lead research-object packaging, metadata, licensing, persistent identifiers, repository integration, and scholarly-record connections through OPE. The approved budget provides one month of effort per year for each of these four senior personnel.

The team builds on continuous Beocat HPC and user support, Sunderland innovation and instructional programming, and OPE repository, publishing, rights, metadata, and public-access services [20, 26, 25]. The award adds a coordinator, administrative assistant, two software engineers, DevOps specialist, Sunderland instructor, and two graduate researchers; NABC, OEIE, and external experts provide security, independent evaluation, and bounded readiness review.

5.2 Project Organization and Timeline

The PI is accountable to NSF, chairs the leadership team, and has final decision authority. Co-PIs report to the PI for their work packages; the coordinator reports to the PI and manages schedule, intake, reporting, and cross-unit follow-through. K-State Libraries is the lead department and manages fiscal, procurement, repository, sponsor-facing, and administrative activity. For Beocat purchases, Engineering and Beocat define specifications and acceptance tests while Libraries manages the award workflow.

The coordinator, administrative assistant, and graduate researchers report to Molik; the software engineers and DevOps specialist to Eggleston; and the instructor to Sheldon. Beocat holds technical release authority, OPE holds repository and metadata acceptance authority, Sunderland holds instructional-readiness authority, NABC and designated university offices retain their security authorities, and OEIE preserves evaluation independence. The Advisory Committee advises but has no line authority. Leadership meets biweekly in Year 1 and monthly thereafter. Quarterly advisory seats represent NABC, CSE/Beocat, KWI and research administration, OEIE, the Kansas Agricultural Experiment Station, and GPN; the PI and co-PIs attend all meetings, and one meeting each year includes the formal strategic and performance review. Kate Adams will serve as the GPN technical collaborator supporting implementation and external engagement.

WBS	Work package	Lead and budget basis	Primary deliverable / acceptance evidence
1	Leadership, coordination, and administration	PI; coordinator; administrative assistant; personnel and travel	PI-led reporting chain, integrated schedule, risk register, quarterly advisory review, NSF reporting
2	Registry, APIs, and software integration	Beocat engineers and DevOps; OPE; personnel and OpenStack equipment	OCI-conformant registry, APIs, CI/CD examples, identity, monitoring, tested releases
3	Execution, storage, and continuity	Beocat; equipment, storage, cloud, network, HA	Managed execution, active storage, recovery exercises, operational targets
4	Research objects and repository integration	OPE and graduate researchers; personnel and storage	FAIR package profile, persistent records, K-REx and external exchange
5	Training and heterogeneous validation	Sunderland instructor and staff; personnel, materials, testbed, participant support	Jetson/edge-AI curriculum, workshops, heterogeneous validation cases
6	Security and trustworthiness	NABC and Beocat; other direct costs	Risk assessments, controls, training, incident procedures, annual assessment
7	Independent evaluation	OEIE; other direct costs	Baseline, readiness, annual, summative, and public evaluation reports
8	National engagement and sustainability	PI/co-PIs, coordinator, GPN; travel and participant support	OAK/GPN and national cohorts, workshops, transferability and sustainability plan

Quarter	Stage	Milestone and acceptance evidence
Y1 Q1	Mobilization	Staff recruitment; baselines; pilot selection; architecture and procurement approval; initial access, security, and data policies
Y1 Q2	Deployment	Registry, API, identity, monitoring, Beocat execution, storage, and K-REx integration in staging; closed K-State tests

Quarter	Stage	Milestone and acceptance evidence
Y1 Q3	Early operations	Invited OAK/GPN beta; GitHub/GitLab and OCI tests; external import/export profile; security and recovery exercises; corrective actions
Y1 Q4	Acceptance	Independent readiness review; policies and documentation published; targets achieved; critical findings closed; full operations by Month 12
Y2 Q1	Expansion	First post-acceptance review; quotas and support refined; two additional pilots and first Year 2 workshop
Y2 Q2	Interoperability	Additional repository, workflow, transfer, or data-access profiles; heterogeneous validation expansion
Y2 Q3	National operations	Broader national cohort; second and third workshops; annual security and continuity exercises
Y2 Q4	Performance review	Annual evaluation and public metrics; sustainability and renewal decision package
Y3 Q1	Scale	Additional national institutions and workflows; transferability testing with partner staff
Y3 Q2	Federation readiness	Node-onboarding and operating guidance; external adoption or emulation exercise
Y3 Q3	Sustainability transition	Final cost, governance, staffing, and partnership model; fourth annual workshop cycle
Y3 Q4	Closeout and continuity	Summative and public evaluation; preservation and operations handoff; final reporting and renewal transition if invited

5.3 Outsourcing

K-State procurement and contract-review processes will govern acquisitions and professional services. Solicitations will specify performance, warranty, support, delivery, security, and acceptance requirements; cloud services will use approved agreements, access and cost controls, export procedures, and Section 4 recovery targets.

The approximately $150,000-per-year professional-services line will be competitively procured. Permissible market research and nonbinding quotes may refine requirements but will not replace competition. Statements of work will assign a university owner and define schedule, acceptance criteria, confidentiality, remedies, and bounded architecture, high-availability, recovery, security-control, certification-readiness, sustainability, and transferability deliverables. Consultants will advise and validate without operational authority or duplication of university work. Early ordering, staged deployment, conformant alternatives, and temporary use of existing or cloud capacity will mitigate delay.

5.4 Software

FORGE will adopt mature infrastructure components and develop only the integration and verification functions unique to the service. K-State Libraries' current cloud-hosted InvenioRDM test instance will provide the starting environment for a maintained production deployment, using a fork where upstream extension points are insufficient, to support the workflow record, DataCite DOI landing page, versioning, access control, search, and repository functions. An established OCI Distribution-compatible registry, such as Harbor or CNCF Distribution, will store and distribute container artifacts. FORGE will add the `Workflow` resource profile, OCI external-artifact adapter, registry and execution connectors, intake and policy workflows, provenance and validation capture, verification-package export, public metrics, and maintained GitHub Actions and GitLab CI/CD templates. Existing Slurm, Kubernetes, OpenStack, Ganglia, institutional identity, database, and S3-compatible storage technologies will be configured and extended rather than replaced.

The primary FORGE-authored software, including the InvenioRDM workflow profile and OCI adapter, intake and policy workflows, execution connectors, verification-package services, and public APIs, will be released under the Mozilla Public License 2.0 (SPDX identifier `MPL-2.0`) in public version control [30]. This file-level copyleft license will allow institutions and vendors to integrate

FORGE components with differently licensed systems while requiring modifications to covered FORGE files to remain available under the MPL. The InvenioRDM deployment, any maintained fork, and inherited dependencies will retain their applicable upstream licenses; FORGE repositories will include machine-readable licensing and dependency information and will contribute generally useful changes upstream when practical.

Software engineers lead application and connector development; DevOps leads deployment, monitoring, release engineering, and recovery; Beocat, OPE, Sunderland, and graduate researchers provide architecture, repository, rights, testing, and user-facing expertise. Releases require review, automated tests, security checks, staging validation, documentation, and rollback readiness. Commercial software requires a documented operational and total-cost justification.

The Data Management and Sharing Plan organizes outputs as: (1) software, APIs, and documentation; (2) executable research objects and registry metadata; (3) benchmark, validation, and operational data; and (4) training and evaluation products. Releases will use the standards specified above, including Git and Semantic Versioning, OCI and OpenAPI, DataCite and ORCID, RO-Crate, W3C PROV, checksums, and machine-readable license and dependency records. Non-sensitive products will be released on a rolling basis and archived in K-REx or another persistent repository, with significant products retained for at least ten years. Credentials, vulnerabilities, sensitive topology, identifiable logs, and identifiable participant records will remain protected. Eggleston is accountable for software and operational data, Molik for workflow records and registry metadata, and Sheldon for training products; OEIE remains responsible for evaluation methods, privacy review, and reporting.

5.5 Risk Management

Risk	Prob.	Impact	Owner / trigger	Mitigation and contingency
Baseline definitions differ across precursor services	Medium	High	PI; inconsistent measures at Q1 review	Apply common definitions to container, repository, Jetson, Ganglia, staffing, and user evidence; audit baselines; prioritize named pilots; report limitations consistently
Year 1 procurement, hiring, or InvenioRDM/OCI integration delay	Medium	High	Coordinator; milestone slips >30 days	Early recruitment and ordering; prototype artifact adapter in Q1; digest-link fallback; staged deployment; existing-capacity or cloud bridge; monthly recovery
Insufficient external adoption	Medium	High	PI/GPN; cohort or workshop target missed by 20%	K-State-to-OAK/GPN-to-national sequence; quarterly outreach; reusable remote onboarding; revise support burden and recruitment channels
Container supply-chain or data-security incident	Medium	High	Beocat/NABC; critical vulnerability or incident	Immutable digests, scoped credentials, scanning, attestations, logging, isolation, incident response, notification, recovery, and post-incident review
Facilities, GovCloud, Nautilus, or storage constraint	Medium	High	PI/Beocat; cost, policy, or availability exceeds threshold	Limit award costs to attributable components; monthly review; local execution remains primary; external endpoints are noncritical; alternative conformant service
Post-award service cannot be sustained	Medium	High	PI; no approved model by Y2 Q4	Preserve K-State core, develop OAK/GPN and national partnership options, quantify cost per service level, seek renewal, consortium support, or transparent intensive-use recovery

The coordinator will maintain the detailed risk register with probability, impact, owner, trigger,

mitigation, contingency, due date, and status. The leadership team will review it monthly in Year 1 and quarterly thereafter; the Advisory Committee will review the highest risks at each quarterly meeting.

5.6 Security and Trustworthiness

FORGE will follow applicable K-State information-security, research-security, acceptable-use, data-governance, physical-access, records, and incident-response policies, informed by the NIST Cybersecurity Framework and OCI software-supply-chain practice. The PI is accountable; Beocat implements operational controls; NABC leads recurring physical, data, and research-security assessment; OPE manages rights, access, metadata, and preservation boundaries; the coordinator maintains training and incident records; and institutional authorities retain their assigned responsibilities.

Assessment will occur during architecture review, before sensitive or controlled workflows, after material changes, and at least annually. Controls include federated authentication, least privilege, scoped credentials, immutable digests, workload isolation, appropriate encryption, monitoring, dependency and vulnerability review, backup and recovery, retention and revocation, separation of duties, change control, training, tabletop exercises, vendor requirements, physical access, asset inventory, and environmental monitoring.

Incidents will be contained, investigated, escalated, and reported under institutional and award requirements, with affected users and partners notified as appropriate. NABC will assess the program annually; OEIE will assess user understanding; and external review will test architecture, AWS GovCloud recovery, supply-chain controls, and certification readiness. FORGE will map controls to a recognized framework and pursue feasible assurance milestones without claiming that a provider authorization certifies the service. Public TRUST reporting will describe scope, responsibilities, aggregate incidents, limitations, continuity evidence, and corrective actions without exposing protected information.

6. Budget Estimation

FORGE requests $9,000,000 over three years, including $6,422,965 in direct costs and $2,577,035 in indirect costs, to integrate and operate the regional service base as a nationally accessible verification service. K-State has approved the internal proposal budget.

Cost category	Year 1	Year 2	Year 3	Total
Salaries, wages, and fringe benefits	$778,329	$801,679	$825,727	$2,405,735
Equipment	$1,450,000	$0	$0	$1,450,000
Domestic travel	$36,000	$37,980	$40,069	$114,049
Participant support	$14,000	$14,620	$15,269	$43,889
Other direct costs	$1,088,768	$642,567	$677,957	$2,409,292
Total direct costs	**$3,367,097**	**$1,496,846**	**$1,559,022**	**$6,422,965**
Indirect costs	$997,475	$773,799	$805,761	$2,577,035
Total project costs	**$4,364,572**	**$2,270,645**	**$2,364,783**	**$9,000,000**

Year 1 is front-loaded for full operations by Month 12; Years 2 and 3 support continuous service, national onboarding, interoperability, evaluation, and sustainability.

Year 1 equipment comprises a $400,000 edge-AI reproducibility testbed, $750,000 Beocat Object Storage expansion, and $300,000 OpenStack environment for heterogeneous validation, active staging, and persistent repository and application services. Libraries will lead acquisition with Engineering and Beocat; the Detailed Cost Estimate provides the line-item basis.

The $500,000 high-availability allowance uses a second UPS on the existing data-center feed and project-specific downstream distribution, monitoring, installation, testing, and commissioning as its planning basis. Facilities and Beocat will confirm the final design and value-engineer the work within

the approved ceiling. The request excludes general renovation, unrelated deferred maintenance, and expansion of campus utility capacity. Equipment selection will remain within the data center's supportable power and cooling envelope. A second network path and limited AWS GovCloud warm environment support recovery without duplicating production.

OEIE provides independent evaluation and NABC recurring security review. The $443,524 professional-services line (approximately $150,000 annually) for bounded architecture, recovery, assurance, and transferability work will be competitively procured; nonbinding market research will only refine requirements.

Other funding and transition. Institutional support continues the campus missions of Beocat, OPE/K-REx, and Sunderland; IDSS supports FORGE integration and national operations. Ag-DEx addresses agricultural data management and governance, while FORGE addresses workflow publication, verification, DataCite records, and run evidence. Any shared pilot will use separate effort, infrastructure, deliverables, accounts, and work packages. No current award funds FORGE, and institutional operations are not voluntary committed cost sharing.

Domestic travel covers the annual NSF PI meeting or designated conference and necessary advisory and GPN/OAK engagement. The Detailed Cost Estimate aligns all costs with the WBS and schedule.

7. Broader Impacts

K-FORGE will make managed workflow verification available to institutions that cannot operate comparable infrastructure. It extends DataCite records, versioning, licensing, contributor attribution, and scholarly relationships to computational workflows, adding durable execution evidence for regional universities, primarily undergraduate institutions, repositories, publishers, and research communities using national infrastructure.

Two OPE graduate researchers will form the initial cohort of K-State Libraries' Digital Scholarship Training Program. Molik will provide weekly supervision, Individual Development Plans, quarterly reviews, and a monthly seminar. Sibley will mentor data stewardship; Eggleston and Beocat staff will provide experience in execution, containers, reliability, and cybersecurity; and Sheldon and the Sunderland instructor will guide progression from observation to co-teaching and supervised delivery. Assignments will include workflow records, pilot documentation, validation, repository integration, training resources, and stakeholder feedback, with preparation in responsible conduct, privacy, licensing, accessibility, and attribution.

K-FORGE will extend training beyond stand-alone workshops by embedding reproducible computational practice within existing courses. The Sunderland Innovation Lab instructor will develop online modules addressing workflow packaging, containers, provenance, persistent identification, research objects, managed execution, and heterogeneous computing. Working with course instructors, the project will incorporate appropriate modules, demonstrations, and assignments into established courses, allowing students to use K-FORGE concepts and tools in disciplinary and technical contexts they already encounter. The modules will be piloted alongside the project's initial workflows, revised using learner and instructor feedback, and updated as the production service evolves. This course-connected approach will broaden access to K-FORGE training, create a recurring pathway for workforce preparation, and produce instructional materials and evidence that can support adoption by other institutions.

Great Plains Network workshops will extend this training and adoption model nationally. Public software, workflows, verification profiles, curricula, implementation guidance, sanitized evaluation products, and the public evaluation report will be released in accessible formats and preserved through K-REx or another recognized repository.

8. Intellectual Merit

FORGE asks what is required to make a computational workflow a citable, executable, and verifiable part of the scholarly record. It will test how a DataCite workflow record, immutable OCI artifact, provenance package, and managed verification service can be joined, and how FAIR4RS and TRUST become measurable requirements [3, 29]. It will identify which DOI, digest, source,

build, provenance, and verification relationships remain portable across registries, repositories, and computing environments.

9. Results from Prior NSF Support

David C. Molik

NSF Award 2126334, "RCN: Reimagining a Sustainable Data Network to Accelerate Agricultural Research and Discovery," $1,238,439, August 2021–August 2026. Phoenix Bioinformatics is the prime recipient. Molik participates through Kansas State University subaward 2126334-005 ($95,729, January–August 2026; 0.75 person-month) and serves on the AgBioData Steering Committee.

Intellectual Merit. AgBioData coordinates agricultural genetics, genomics, and breeding databases around FAIR practices, interoperable standards, publication-to-database pathways, and sustainability. Molik contributed to publisher, library, and database coordination; a specification and open-source tool for consistent genome-assembly and annotation nomenclature [5]; and the AgBioDatabase Finder, which directs authors to appropriate disciplinary repositories [4]. These products address inconsistent identifiers and metadata, fragmented repository guidance, and weak linkages among data, publications, and community databases.

Broader Impacts. AgBioData engages database teams, researchers, curators, publishers, librarians, students, and government scientists through workshops, webinars, implementation guidance, and a seven-module FAIR agricultural-data curriculum. These activities strengthen data-stewardship skills and improve the visibility, reuse, and sustainability of publicly funded agricultural information resources.

Research products. Relevant publications include *Guidelines for Gene and Genome Assembly Nomenclature* [5] and *AgBioDatabase Finder: an Online Tool to Help Researchers Find and Submit Agricultural Genomic, Genetic, and Breeding Data* [4]. Additional products include the open-source Assembly Name Generator, AgBioDatabase Finder and source dataset, FAIR curriculum, community recommendations, user guides, and archived training materials. Products are publicly available through AgBioData, GitHub, Zenodo, CaltechDATA, NSF-PAR, and open-access journals.

Relationship to the proposed work. This support informs FORGE's community governance, FAIR metadata, persistent identification, repository coordination, training, and sustainability. It does not overlap scientifically or financially with FORGE. AgBioData develops standards and practices for agricultural databases; FORGE will operate a cross-disciplinary service for publishing, executing, and verifying computational workflows and linking DataCite records to immutable OCI artifacts and execution evidence.

Nicholas Eggleston

No Prior NSF Support. Nicholas Eggleston has no prior NSF support to report under the NSF Proposal and Award Policies and Procedures Guide, including no current NSF funding and no NSF award with an end date within the five years preceding this proposal.

Jeffrey Sheldon

No Prior NSF Support. Jeffrey Sheldon has no prior NSF support to report under the NSF Proposal and Award Policies and Procedures Guide, including no current NSF funding and no NSF award with an end date within the five years preceding this proposal.

Gwen Sibley

No Prior NSF Support. Gwen Sibley has no prior NSF support to report under the NSF Proposal and Award Policies and Procedures Guide, including no current NSF funding and no NSF award with an end date within the five years preceding this proposal.

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SUMMARY PROPOSAL BUDGET

YEAR 1

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months) Proposed	Granted
2638522		
AWARD NO.		

ORGANIZATION: Kansas State University

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: David Molik

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. David Molik - Principal Inv	1.0			9,013	
2. Nicholas Eggleston	1.0			10,729	
3. Jeffrey Sheldon	1.0			9,013	
4. Gwendolyn Sibley	1.0			6,867	
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (4) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	4.0			35,622	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (3) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	12.0			435,900	
3. (2) GRADUATE STUDENTS				70,500	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (1) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				49,000	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				591,022	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				187,307	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				778,329	

D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)

Edge-AI reproducibility testbed	$ 400000
Repository and workflow storage upgrade, or Beocat Object Storag	$ 750000
OpenStack repository and application service environment upgrade	$ 300000

TOTAL EQUIPMENT	1,450,000	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)	36,000	
2. INTERNATIONAL	0	

F. PARTICIPANT SUPPORT COSTS

1. STIPENDS $	0
2. TRAVEL	6,000
3. SUBSISTENCE	0
4. OTHER	8,000

TOTAL NUMBER OF PARTICIPANTS (106) TOTAL PARTICIPANT COSTS	14,000	
G. OTHER DIRECT COSTS		
1. MATERIALS AND SUPPLIES	52,700	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION	12,000	
3. CONSULTANT SERVICES	140,000	
4. COMPUTER SERVICES	98,000	
5. SUBAWARDS	0	
6. OTHER	786,068	
TOTAL OTHER DIRECT COSTS	1,088,768	
H. TOTAL DIRECT COSTS (A THROUGH G)	3,367,097	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 53.0, Base:1882029)		
TOTAL INDIRECT COSTS (F&A)	997,475	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)	4,364,572	
K. FEE	0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)	4,364,572	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: David Molik

ORG. REP. NAME*: Bailey J Starns

FOR NSF USE ONLY		
INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 2

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months)	
	Proposed	Granted
AWARD NO.		

ORGANIZATION: Kansas State University

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: David Molik

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. David Molik - Principal Inv	1.0			9,283	
2. Nicholas Eggleston	1.0			11,051	
3. Jeffrey Sheldon	1.0			9,283	
4. Gwendolyn Sibley	1.0			7,073	
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (4) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	4.0			36,690	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (3) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	12.0			448,977	
3. (2) GRADUATE STUDENTS				72,615	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (1) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				50,470	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				608,752	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				192,927	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				801,679	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
Edge-AI reproducibility testbed $ 0					
Repository and workflow storage upgrade, or Beocat Object Storag $ 0					
OpenStack repository and application service environment upgrade $ 0					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				37,980	
2. INTERNATIONAL				0	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 0					
2. TRAVEL 6,180					
3. SUBSISTENCE 0					
4. OTHER 8,440					
TOTAL NUMBER OF PARTICIPANTS (106) TOTAL PARTICIPANT COSTS				14,620	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				12,660	
3. CONSULTANT SERVICES				147,700	
4. COMPUTER SERVICES				206,780	
5. SUBAWARDS				0	
6. OTHER				275,427	
TOTAL OTHER DIRECT COSTS				642,567	
H. TOTAL DIRECT COSTS (A THROUGH G)				1,496,846	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 53.0, Base:1459999)					
TOTAL INDIRECT COSTS (F&A)				773,799	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				2,270,645	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				2,270,645	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: David Molik

ORG. REP. NAME*: Bailey J Starns

FOR NSF USE ONLY		
INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 3

FOR NSF USE ONLY		
PROPOSAL NO.	DURATION (months) Proposed	Granted
2638522		
AWARD NO.		

ORGANIZATION: Kansas State University

PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: David Molik

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. David Molik - Principal Inv	1.0			9,561	
2. Nicholas Eggleston	1.0			11,383	
3. Jeffrey Sheldon	1.0			9,561	
4. Gwendolyn Sibley	1.0			7,285	
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)	0.0			0	
7. (4) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	4.0			37,790	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (3) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	12.0			462,446	
3. (2) GRADUATE STUDENTS				74,793	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (1) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				51,984	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				627,013	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				198,714	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				825,727	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
Edge-AI reproducibility testbed $ 0					
Repository and workflow storage upgrade, or Beocat Object Storag $ 0					
OpenStack repository and application service environment upgrade $ 0					
TOTAL EQUIPMENT				0	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				40,069	
2. INTERNATIONAL				0	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 0					
2. TRAVEL 6,365					
3. SUBSISTENCE 0					
4. OTHER 8,904					
TOTAL NUMBER OF PARTICIPANTS (106) TOTAL PARTICIPANT COSTS				15,269	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				0	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				13,356	
3. CONSULTANT SERVICES				155,824	
4. COMPUTER SERVICES				218,202	
5. SUBAWARDS				0	
6. OTHER				290,575	
TOTAL OTHER DIRECT COSTS				677,957	
H. TOTAL DIRECT COSTS (A THROUGH G)				1,559,022	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 53.0, Base:1520303)					
TOTAL INDIRECT COSTS (F&A)				805,761	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				2,364,783	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				2,364,783	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME: David Molik

ORG. REP. NAME*: Bailey J Starns

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INDIRECT COST RATE VERIFICATION		
Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION	PROPOSAL NO.	DURATION (months)	
Kansas State University	2638522	Proposed	Granted
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR: David Molik	AWARD NO.		

FOR NSF USE ONLY

A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)	NSF Funded Person-months CAL	ACAD	SUMR	Funds Requested By proposer	Funds granted by NSF (if different)
1. David Molik - Principal Inv	3.0			27,857	
2. Nicholas Eggleston	3.0			33,163	
3. Jeffrey Sheldon	3.0			27,857	
4. Gwendolyn Sibley	3.0			21,225	
5.					
6. ( ) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)					
7. (4) TOTAL SENIOR/KEY PERSONNEL (1 - 6)	12.0			110,102	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)					
1. (0) POST DOCTORAL SCHOLARS	0.0			0	
2. (9) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)	36.0			1,347,323	
3. (6) GRADUATE STUDENTS				217,908	
4. (0) UNDERGRADUATE STUDENTS				0	
5. (3) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)				151,454	
6. (0) OTHER				0	
TOTAL SALARIES AND WAGES (A + B)				1,826,787	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)				578,948	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)				2,405,735	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING $10,000.)					
TOTAL EQUIPMENT				1,450,000	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)				114,049	
2. INTERNATIONAL				0	
F. PARTICIPANT SUPPORT COSTS					
1. STIPENDS $ 0					
2. TRAVEL 18,545					
3. SUBSISTENCE 0					
4. OTHER 25,344					
TOTAL NUMBER OF PARTICIPANTS (318) TOTAL PARTICIPANT COSTS				43,889	
G. OTHER DIRECT COSTS					
1. MATERIALS AND SUPPLIES				52,700	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION				38,016	
3. CONSULTANT SERVICES				443,524	
4. COMPUTER SERVICES				522,982	
5. SUBAWARDS				0	
6. OTHER				1,352,070	
TOTAL OTHER DIRECT COSTS				2,409,292	
H. TOTAL DIRECT COSTS (A THROUGH G)				6,422,965	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)					
TOTAL INDIRECT COSTS (F&A)				2,577,035	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)				9,000,000	
K. FEE				0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)				9,000,000	

M. COST SHARING PROPOSED LEVEL $ 0 AGREED LEVEL IF DIFFERENT $

PI/PD NAME	FOR NSF USE ONLY		
David Molik	INDIRECT COST RATE VERIFICATION		
ORG. REP. NAME*	Date Checked	Date Of Rate Sheet	Initials - ORG
Bailey J Starns			

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

BUDGET JUSTIFICATION

Category II: K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science

Kansas State University defines year as academic year

Project Period: January 1, 2027–December 31, 2029 | Total Request: $9,000,000

Kansas State University requests $9,000,000 for the three-year K-FORGE project period. The request includes $6,422,965 in direct costs and $2,577,035 in facilities and administrative costs. The budget supports integration of established regional research-computing, repository, workflow-support, security, and instructional capabilities into an operational service by the end of Year 1, followed by national user operations, service improvement, evaluation, and sustainability activities in Years 2 and 3. Salaries are escalated at 3 percent annually; graduate tuition and selected operating costs are escalated at 5.5 percent annually. The separate Detailed Cost Estimate maps the major infrastructure and service costs to work packages, quantities, planning assumptions, and project years. This justification explains the necessity and reasonableness of each budget category while preserving appropriate procurement flexibility.

A. SENIOR/KEY PERSONNEL

David Molik, Principal Investigator

The budget requests one month of effort in each project year for David Molik ($9,013 in Year 1, $9,283 in Year 2, and $9,561 in Year 3; $27,857 total salary). Molik will provide overall project leadership; integrate the research-computing, repository, scholarly-communication, evaluation, and community-engagement activities; oversee governance and sponsor communication; coordinate milestones; and lead dissemination and sustainability planning.

Nicholas Eggleston, Co-Principal Investigator

The budget requests one month of effort in each project year for Nicholas Eggleston ($10,729 in Year 1, $11,051 in Year 2, and $11,383 in Year 3; $33,163 total salary). Eggleston will lead production research computing and cyberinfrastructure, including Beocat integration, systems architecture, managed execution, storage, scheduling, infrastructure security, monitoring, reliability, and operational readiness.

Jeffrey Sheldon, Co-Principal Investigator

The budget requests one month of effort in each project year for Jeffrey Sheldon ($9,013 in Year 1, $9,283 in Year 2, and $9,561 in Year 3; $27,857 total salary). Sheldon will lead Sunderland Innovation Lab and AI Studio activities, including workflow prototyping, instructional integration, student engagement, demonstrations, heterogeneous-computing activities, documentation, and translation of workflows into production-ready forms.

Gwendolyn Sibley, Senior/Key Personnel

The budget requests one month of effort in each project year for Gwendolyn Sibley ($6,867 in Year 1, $7,073 in Year 2, and $7,285 in Year 3; $21,225 total salary). Sibley will support registry and metadata design, research-object packaging, repository integration, persistent identifiers, licensing, citation metadata, documentation, researcher onboarding, and connections among workflows, run records, and durable scholarly outputs.

Total senior/key personnel salary is $110,102. Associated fringe benefits total $36,334, producing total senior/key personnel compensation of $146,436.

B. OTHER PERSONNEL

Sunderland Instructor and Project Coordinator

The budget provides $406,453 over three years ($131,500 in Year 1, $135,445 in Year 2, and $139,508 in Year 3) for an instructor supporting Sunderland Innovation Lab and AI Studio activities and a project coordinator supporting the overall K-FORGE service. The instructor will support workflow prototyping, student supervision, training, documentation, and demonstrations. The project coordinator will manage

schedules, milestones, onboarding, reporting, meetings, project records, risk tracking, user-support coordination, and dependencies among project units.

Research Software Engineers and DevOps Personnel
The budget provides $940,870 over three years ($304,400 in Year 1, $313,532 in Year 2, and $322,938 in Year 3) for two research software engineers and one DevOps/cyberinfrastructure professional. These personnel will implement and maintain the registry, APIs, workflow packaging and automation, execution connectors, deployment processes, monitoring, identity and authorization integrations, metadata exchange, repository handoffs, testing, and production operations.

Graduate Research Assistants
The budget provides $217,908 in stipends for two graduate research assistants ($70,500 in Year 1, $72,615 in Year 2, and $74,793 in Year 3). The graduate research assistants will contribute to workflow translation, test development, provenance and metadata validation, benchmarking, documentation, pilot support, data analysis, and evaluation-related technical work under project supervision.

Administrative Assistant
The budget provides $151,454 over three years ($49,000 in Year 1, $50,470 in Year 2, and $51,984 in Year 3) for administrative support. This position will assist with purchasing, travel, meeting logistics, records, participant communications, documentation routing, and coordination of project administrative requirements.

Other-personnel salaries total $1,716,685. Applicable fringe benefits total $542,614, producing total other-personnel compensation of $2,259,299.

C. FRINGE BENEFITS
Fringe benefits are calculated using Kansas State University rates applicable to each appointment category. The budget includes $578,948 in total fringe benefits: $187,307 in Year 1, $192,927 in Year 2, and $198,714 in Year 3.

D. EQUIPMENT
The budget requests $1,450,000 in Year 1 for equipment necessary to establish the operational K-FORGE service environment. The estimate is divided among a $400,000 low-power edge-AI reproducibility testbed, a $750,000 repository and workflow storage upgrade, and $300,000 in OpenStack repository and application service nodes. These components support distinct but integrated functions: heterogeneous workflow validation, high-throughput active data and artifact storage, and reliable hosting of registry and application services. Purchases will be competitively procured and coordinated with Beocat, university information-technology staff, facilities personnel, and Sponsored Programs. Planning estimates include the primary systems, required memory and storage, networking and optics, rack integration, power components, remote management, warranty or support, and reasonable spare capacity. Final makes, models, capacities, and component counts may be adjusted during procurement while preserving the functions and category totals described below.

Low-Power Edge-AI Reproducibility Testbed — $400,000
The $400,000 estimate will support a 24-node, rack-integrated low-power testbed, anticipated as approximately six four-node chassis or an equivalent serviceable configuration. The current planning model allocates eight nodes to each of three representative accelerator classes - TPU, FPGA, and NPU - with approximately four M.2 accelerator positions per node, providing 96 total accelerator positions. This arrangement allows the same packaged workflow to be executed across multiple low-power processor and accelerator environments under controlled conditions. The estimate includes the chassis and node modules; representative accelerator devices; memory; local redundant storage for operating systems, workflow staging, logs, and telemetry; a management or head node; switching, network adapters, and optics; rack rails, power distribution, and cabling; remote-management capability; spare components; and vendor warranty or support. Final accelerator selections will emphasize devices with stable software interfaces and sufficient community relevance to support reproducibility testing. The testbed will evaluate

compatibility, portability, performance variation, telemetry capture, and reproducibility across emerging accelerator environments. It is a supporting component of the workflow execution service, not a stand-alone general-purpose AI instrument.

Repository and Workflow Storage Upgrade — $750,000
The $750,000 estimate will provide production storage capacity for active workflow staging, large reference datasets, software and container artifacts, workflow outputs, logs, run-level provenance, benchmark artifacts, repository ingest, backup, and handoff to public-access systems. The planning estimate separates the storage function into enterprise storage servers or enclosures; high-capacity media for usable data capacity; NVMe or comparable cache, metadata, and high-input/output tiers; redundant controllers and power supplies; high-throughput network interfaces, switches, and optics; rack integration and cabling; replication or backup components; spare media; and vendor warranty and support. The design must accommodate both large sequential research data transfers and the high volume of smaller files, metadata records, logs, and provenance objects generated by workflow execution. Final capacity and architecture will be selected through technical design and competitive procurement to balance usable capacity, throughput, redundancy, rebuild time, maintainability, and integration with existing Beocat and repository storage. This equipment supports active project operations and repository staging; it is not intended to obligate NSF to indefinite preservation of all project-generated data.

OpenStack Repository and Application Service Nodes — $300,000
The $300,000 estimate will provide a redundant set of production-grade OpenStack compute and application nodes for the registry, repository, API, identity, metadata, monitoring, database, and supporting K-FORGE services. The estimate includes server chassis, processors, memory, mirrored or redundant local storage, network interfaces and optics, remote-management capabilities, rack integration, spare components, and vendor warranty or support. The configuration will provide sufficient capacity to separate critical control, application, database, identity, and monitoring roles and to retain node-level redundancy during planned maintenance or isolated failures. These nodes will host persistent service functions independently from transient research workloads on Beocat, allowing the execution environment and the public-facing registry and repository interfaces to scale and be maintained on different operational schedules. Final node counts and specifications will be selected to meet service-readiness, high-availability, and lifecycle-support requirements within the approved category total.

E. TRAVEL

Domestic Travel — $114,049
The budget requests $36,000 in Year 1, $37,980 in Year 2, and $40,069 in Year 3 for domestic travel. Travel will support required NSF meetings, project-team coordination, community and pilot engagement, technical collaboration, dissemination at relevant research-cyberinfrastructure and open-science conferences, and limited site or partner visits needed to operate and evaluate the service. No foreign travel is requested.

F. PARTICIPANT SUPPORT COSTS

The budget requests $43,889 in participant support costs: $14,000 in Year 1, $14,620 in Year 2, and $15,269 in Year 3. This amount includes $18,545 for travel by eligible non-employee advisory, community, and workshop participants and $25,344 for participant meeting support associated with project workshops, demonstrations, and advisory activities. Participant support will not be used for employee compensation or routine project operations.

G. OTHER DIRECT COSTS

Materials and Supplies — $52,700
Year 1 materials and supplies include $24,000 for workstations used by project personnel, $25,200 for the Sunderland instructional and workflow-prototyping compute laboratory and representative accelerator devices, and $3,500 for PI computing equipment. These items fall below the institutional equipment

threshold and will support software development, documentation, workflow testing, training, and project administration.

Publication, Documentation, and Dissemination — $38,016
The budget requests $12,000 in Year 1, $12,660 in Year 2, and $13,356 in Year 3 for publication and dissemination. Costs may include open-access publication charges, preparation and hosting of public technical documentation, production of training and implementation materials, and dissemination of the public evaluation report and reusable project outputs.

Consultant and Professional Services — $443,524
The budget requests $140,000 in Year 1, $147,700 in Year 2, and $155,824 in Year 3 for specialized external review and professional services. Anticipated services include independent service-readiness validation, high-availability architecture review, operational and sustainability analysis, transferability review, and specialized engineering or security consultation not available within the project team or participating institutional units. Expected deliverables may include architecture and readiness assessments, prioritized remediation recommendations, review of recovery and continuity procedures, validation of operating assumptions, sustainability and cost-model analysis, and an external assessment of whether the implementation guidance can be transferred to institutions with different technical capacities. These services complement, rather than duplicate, the physical and data security work assigned to the National Agricultural Biosecurity Center or the operational implementation performed by Beocat personnel. Specific scopes, rates, periods of performance, and deliverables will be established through university procurement and contracting procedures. No consultant will exercise project direction or perform duties assigned to university employees.

Computer Services — $522,982
The budget requests $98,000 in Year 1, $206,780 in Year 2, and $218,202 in Year 3 for cloud computing and storage supporting a warm continuity environment for critical K-FORGE services, for a total of $522,982,. Year 1 reflects a partial-year deployment and configuration period; Years 2 and 3 support full-year operations. Anticipated services include compute capacity for selected registry and application functions, replicated object or block storage, snapshots and backup, network traffic, monitoring, logging, security services, and limited failover or recovery exercises. The cloud environment will not duplicate the full Beocat execution platform. Instead, it will preserve selected critical service functions and data needed to restore or continue operations following a local outage, maintenance event, or infrastructure failure. Usage and cost will be monitored throughout the project, with service sizing adjusted to actual continuity and recovery requirements. The architecture will distinguish production services, continuity functions, active workflow storage, and durable repository preservation.

Graduate Tuition — $66,744
The budget requests $21,068 in Year 1, $22,227 in Year 2, and $23,449 in Year 3 for tuition associated with the two graduate research assistants. Tuition is escalated at 5.5 percent annually.

Office of Educational Innovation and Evaluation — $633,605
The budget requests $200,000 in Year 1, $211,000 in Year 2, and $222,605 in Year 3 for the Office of Educational Innovation and Evaluation (OEIE). OEIE will develop and implement the evaluation framework; establish baselines, performance indicators, and data responsibilities; collect and analyze user, service, and collaboration evidence; administer surveys and structured feedback activities; synthesize advisory committee findings; and provide formative recommendations during implementation and operations. OEIE will prepare annual evaluation products, a summative report, and a public evaluation report documenting service readiness, use, outcomes, limitations, sustainability, and lessons for other institutions. The allocation supports the professional evaluation effort needed to design instruments, manage data collection, conduct analysis, participate in project and advisory meetings, report findings, and translate evidence into actionable recommendations. OEIE's evaluation role is distinct from project management and from the technical performance monitoring conducted by the service team.

Second Beocat Network-Line Upgrade — $25,000
The Year 1 budget includes $25,000 for optics, configuration, testing, and related work needed to activate or upgrade a second network path for Beocat service reliability. The second path will reduce dependence on a single connection for K-FORGE data movement and service operations and will support continuity planning for the registry, repository staging, and managed execution pathways. Scope and costs will be coordinated with campus networking and KanREN before procurement and will be limited to project-attributable components and configuration.

National Agricultural Biosecurity Center Security Support — $126,721
The budget requests $40,000 in Year 1, $42,200 in Year 2, and $44,521 in Year 3 for the National Agricultural Biosecurity Center (NABC) to support physical and data security planning. NABC will conduct research-security review, threat and risk assessment, review of physical-access and asset-protection practices, assessment of workflows involving sensitive or controlled data, development of security and incident-response procedures, and recurring review of controls as the service transitions into national operations. Anticipated products include documented risk and threat assessments, security requirements for workflow onboarding, recommendations for physical and logical controls, incident and escalation playbooks, review findings for selected sensitive-data pilots, and annual reassessment of the service's evolving risk posture. NABC will coordinate with Beocat, university information-security and facilities offices, and independent external reviewers. This allocation establishes a named institutional locus for security governance while preserving separation between operational implementation, institutional authority, and independent review.

Data-Center Power Redundancy and High-Availability Support — $500,000
The budget requests $500,000 in Year 1 for project-specific data-center power redundancy, high-availability support, and associated work required for continuous K-FORGE service operations. The planning amount may support project-attributable electrical and infrastructure engineering; redundant uninterruptible-power, transfer, distribution, or rack-level power components; installation and commissioning; monitoring integration; and related work necessary to reduce single points of failure for the equipment acquired under the project. Final scope will be determined through facilities and data-center engineering review after the production architecture and power requirements are confirmed. Any work will be limited to allowable improvements directly attributable to K-FORGE and will be coordinated with facilities, data-center, engineering, and Sponsored Programs personnel. Funds are not requested for general building renovation, unrelated deferred maintenance, or expansion of campus utility capacity beyond that required for the project.

No subawards are included in the proposed budget.

H. TOTAL DIRECT COSTS

Total direct costs are $3,367,097 in Year 1, $1,496,846 in Year 2, and $1,559,022 in Year 3, for a three-year total of $6,422,965.

I. INDIRECT COSTS

Facilities and administrative costs are calculated using Kansas State University's negotiated 53 percent modified total direct cost rate. The MTDC base excludes equipment and participant support costs, tuition and applies all other applicable exclusions. Indirect costs total $997,475 in Year 1, $773,799 in Year 2, and $805,761 in Year 3, for a three-year total of $2,577,035.

TOTAL REQUEST

The total K-FORGE request is $9,000,000 over three years.

FACILITIES, EQUIPMENT, AND OTHER RESOURCES

Institutional setting. K-FORGE will be led by Kansas State University Libraries through the Open Publishing Exchange (OPE). The important point is not that any one K-State unit already operates the service proposed here. It is that the main pieces are already in place and used in production: a research computing center, repository and publishing programs, a public-facing innovation laboratory, campus networking and information-security support, and domain research facilities that generate the kinds of workflows K-FORGE is intended to register, execute, validate, and connect to the scholarly record. OPE provides the project home because it already works across repository stewardship, open publishing, persistent identifiers, metadata, licensing, public-access compliance, documentation, and researcher support. That is the part of the workflow lifecycle that often falls between research computing and publication, and it is the part K-FORGE is designed to make operational.

Sunderland Foundation Innovation Lab. The Sunderland Foundation Innovation Lab in Hale Library is especially important to the project. Sunderland is not a conventional computer laboratory and will not be treated as a small version of Beocat. It is the place where researchers, students, librarians, technologists, and community users can sit together, take apart a workflow, document what it does, move it into a portable environment, test it on approachable hardware, and decide what is ready for managed execution. The Lab is jointly supported by K-State Libraries and the Division of Information Technology and already operates as an accessible, interdisciplinary environment for experimentation, prototyping, workshops, and technical instruction. It is open to K-State students, faculty, staff, and the broader community, which gives K-FORGE an existing place to test not only whether a workflow runs, but whether its documentation and training are understandable outside the development team. Co-PI Jeffrey Sheldon will lead this part of the work with the project instructor, graduate researchers, and other Sunderland personnel.

Sunderland includes the Artificial Intelligence Studio, collaboration and teaching spaces, equipment checkout, Fabspace and makerspace resources, an immersion studio, Liquid Galaxy, a media studio and gallery, a One Button+ studio, sound and video-production spaces, and loanable cameras and other instructional technology. K-FORGE will use these spaces for workflow intake, supervised prototyping, demonstrations, edge-AI exercises, documentation, accessibility review, and the creation of reusable training materials. The media facilities are particularly useful because they allow the project to turn a successful workshop or troubleshooting session into a reusable tutorial, short demonstration, or onboarding module rather than treating every training event as a one-time activity. The Artificial Intelligence Studio and portable instructional systems will support small, visible examples of model execution, accelerator differences, and resource constraints before comparable workflows are moved into a managed research-computing environment.

The division matters to the project design. A workflow should not arrive at Beocat as an undocumented directory and a request to make it run. Sunderland gives the space to test and work through dependencies, containerization, metadata, licensing, documentation, and teaching before the workflow enters a more controlled service pathway. Graduate researchers will be able to observe an experienced instructor, co-design and co-teach workshops, and then lead supervised sessions as part of the Digital Scholarship Training Program. Researchers can work through a pilot with staff at a table or display, students can reproduce the same steps on modest systems, and the project can record where users become confused before those assumptions are built into production documentation. The project-supported portable edge and instructional systems will extend this environment for workshops and student exercises. They will not duplicate the rack-integrated equipment at Beocat; Sunderland is the human-centered translation, prototyping, and learning environment, while Beocat is the production execution and verification environment.

Beocat research computing. Beocat provides the production computing and data-center environment. It is K-State's established high-performance computing service and the largest academic supercomputer in Kansas, operated by the Institute for Computational Research in Engineering and Science. Its existing environment includes nearly 400 researcher-funded systems, approximately 10,000 processor cores, about 3.3 petabytes of storage, and a heterogeneous GPU

population ranging from older research accelerators to current NVIDIA L40S-class devices. Beocat is available to educational researchers in Kansas and their collaborators, with priority access for researchers who contribute resources. This operating model is directly relevant to K-FORGE because it combines a guaranteed project base with access to a much larger shared environment when validation or benchmark work requires it.

The Beocat environment includes Linux compute resources, Slurm scheduling, GPU and high-memory nodes, OpenStack, Open OnDemand, Globus data transfer, software modules, monitoring, account and allocation practices, and direct research-computing support. Beocat already supports containerized workflows through Apptainer and can connect users to national research-computing resources when a workflow exceeds local capacity. Its staff routinely support research and teaching in areas such as bioinformatics, data science, cybersecurity, deep learning, chemistry, economics, and genetics. That breadth matters because K-FORGE is not proposing a new general-purpose computing center. It is adding a managed verification and repository service to an environment that already knows how to onboard researchers, schedule work, move large datasets, maintain shared systems, document service changes, and help users diagnose why a computational job is not behaving as expected.

Beocat will operate the K-FORGE execution, active workflow storage, registry, and application-service components. The requested equipment will add a bounded heterogeneous edge-AI reproducibility testbed, expand object storage for workflow artifacts, logs, provenance, validation records, and repository staging, and provide OpenStack nodes for InvenioRDM, registry, API, identity, metadata, database, monitoring, and connector functions. These resources will be integrated into existing rack, networking, security, support, and change-management practices rather than operated as a separate experimental island. Project-specific power redundancy, rack distribution, monitoring, and a second network path will reduce single points of failure for K-FORGE equipment. The improvements are limited to the needs of the proposed service and are not a general renovation of the Beocat data center.

K-State's institutional cloud environment will provide geographically separate continuity capacity for selected backups, replicas, monitoring, recovery data, and limited application failover. The cloud environment will not duplicate Beocat's active execution or storage layers. Instead, it will provide a practical warm-recovery path and a separate place to test restore procedures, application deployment, and service continuity. Beocat's Globus support and campus and regional network connections provide the corresponding pathway for moving large research objects and validation packages between K-State, external institutions, and national facilities.

Libraries, repository, and publishing infrastructure. K-State Libraries operates K-REx, the K-State Research Exchange, and New Prairie Press, a diamond open-access publishing program for journals, conference proceedings, monographs, and related scholarly products. The Libraries also maintains a cloud-hosted InvenioRDM test environment. Together, these systems give K-FORGE an existing place to test DataCite workflow records, qualified relationships, repository communities, access controls, object-storage patterns, editorial intake, rights review, and links among publications, datasets, software, exact executable artifacts, provenance, and verification results. OPE staff already work with authors, editors, repository depositors, campus units, and external partners, so the project begins with an established service and communication environment rather than a new technical platform without a user-facing home.

K-FORGE is not simply depositing the output of a computing project in a repository after the work is complete. The repository and publishing systems are part of the service itself. They provide the durable record that says what was run, which version was used, who contributed, what the license permits, where the inputs and outputs are located, and what happened during verification. K-REx provides an institutional preservation and dissemination pathway, while New Prairie Press provides a realistic environment for testing how executable research objects can be connected to journal articles, proceedings, monographs, and other publications. The InvenioRDM environment provides a place to develop and test the more specialized workflow registry and research-object functions before those functions are hardened for production use.

Hale Library workspace and meetings. Hale Library will provide the day-to-day workspace for the PI, OPE personnel, graduate researchers, and project coordination staff. It is a practical project home because OPE, Sunderland, public computing and technology support, the graduate-student hub, and meeting spaces are in the same building. Hale's Grad Hub provides dedicated space and reservable rooms for graduate students, giving the two K-FORGE graduate researchers a place to work independently, meet with mentors, and remain connected to other graduate students outside the project.

Hale has 17 reservable collaboration rooms with whiteboard walls; most include large displays and laptop connections. Additional conference rooms support groups ranging from small project meetings to approximately 30 participants. The Hemisphere Room and the Dr. Susan K. Peterson Room can each accommodate up to 100 people and include built-in presentation systems, projectors, large screens, and Zoom capabilities. K-FORGE will use these spaces for weekly team work, student mentoring, hybrid advisory meetings, design and security reviews, national onboarding, workshops, visiting collaborators, and demonstrations. The point is not that meeting rooms are unusual institutional resources; it is that the project can conduct genuinely hybrid technical work, with remote participants able to see demonstrations and contribute to design discussions, without having to build a separate collaboration facility.

Core facilities and pilot environments. K-State has a useful set of core facilities and domain environments from which K-FORGE can draw realistic workflows. These facilities are not being presented as committed project subunits. They are existing research environments that generate data-intensive, instrument-linked, geospatial, biological, and analytical work with the kinds of dependencies, formats, reference resources, and provenance problems that K-FORGE is intended to address.

The Bioinformatics Center maintains analytical software and pipelines across multiple biological data types, and the Integrated Genomics Facility provides campus-wide sequencing services, technical expertise, research assistance, and training. Together they represent a realistic source of workflows involving large inputs, changing reference databases, specialized software, version-sensitive parameters, and long computational dependency chains. The value of these cases is not simply that they use large data. They give K-FORGE an opportunity to show whether a workflow can be described well enough that another researcher can understand the inputs, reconstruct the environment, and distinguish an expected biological difference from a software or reference-version difference.

The Geographic Information Systems Spatial Analysis Laboratory (GISSAL) is a university core facility for spatiotemporal analytics, geoprocessing-tool development, cartographic and visualization support, GIS web services, map applications, and geospatial data portals. Its GeoCat environment provides a useful cloud-based setting for geospatial storage, processing, visualization, and analysis. A selected geospatial or water-related workflow can therefore be examined in its native GIS environment and in Beocat, giving the project a concrete portability case involving proprietary and open components, large spatial datasets, web services, coordinate and projection assumptions, and visual outputs. The Statistical Consulting Laboratory provides a complementary source of expertise for benchmark design, analysis plans, interpretation of validation results, and the distinction between a technically successful rerun and a scientifically meaningful comparison.

Security, evaluation, and institutional support. The National Agricultural Biosecurity Center will support recurring physical, data, and research-security review, especially for workflows connected to agricultural emergency management, controlled information, or software supply-chain risk. Campus information-security, networking, identity, facilities, legal, compliance, and research-administration offices will continue to provide their normal institutional functions. KanREN connectivity and campus networking provide an established path for regional and national service access, while facilities and data-center personnel provide the local environment for power, rack, cooling, and network work directly attributable to the project.

The Office of Educational Innovation and Evaluation will provide independent evaluation of readiness, usability, performance, adoption, interoperability, transferability, and sustainability. Great Plains Network and OAK relationships provide an existing regional community for workshops, technical

exchange, external users, and dissemination. These relationships are important because K-FORGE has to be understandable and useful outside K-State; the project cannot demonstrate transferability by asking only the people who built the service whether it works.

Overall availability. Together, these resources provide the operational base for K-FORGE. Sunderland provides the place where workflows are made understandable and teachable; Beocat provides managed execution and verification; OPE, K-REx, InvenioRDM, and New Prairie Press connect the computational object to the scholarly record; Hale Library provides the daily workspace and hybrid collaboration environment; and K-State's core facilities, security, evaluation, networking, and regional relationships provide realistic workflows and external review. The requested personnel, equipment, cloud services, and professional review add the capacity needed to connect these existing pieces into a sustained national service.

IDENTIFYING INFORMATION:

NAME: Molik, David

ORCID iD: https://orcid.org/0000-0003-3192-6538

POSITION TITLE: Department Head, Associate Professor, Librarian

PRIMARY ORGANIZATION AND LOCATION: Kansas State University, Mahattan, Kansas, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Notre Dame, Notre Dame, IN, United States	PHD	08/2016 - 06/2021	Integrated Biomedical Sciences, Computational Biology & Bioinformatics
Rensselaer Polytechnic Institute, Troy, NY, United States	BS	08/2009 - 12/2012	Information Technology and Web Science

Appointments and Positions

2025 - present Department Head, Associate Professor, Librarian, Kansas State University, Mahattan, Kansas, United States

2023 - 2025 Computational Biologist - SY - CAT 4, USDA Agricultural Research Service, Arthropod-borne Animal Diseases Research Unit, Manhattan, Kansas, US

2021 - 2022 PostDoc/Research Associate, USDA Agricultural Research Service, Pacific Basin Agricultural Research Center, Hilo, Hawaii, US

2014 - 2016 Scientific Informatics Developer, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, US

Products

Products Most Closely Related to the Proposed Project

1. Wright A, Wilkinson M, Mungall C, Cain S, Richards S, Sternberg P, Provin E, Jacobs J, Geib S, Raciti D, Yook K, Stein L, Molik D. FAIR Header Reference genome: a TRUSTworthy standard. Briefings in Bioinformatics. 2024 May; 25(3):-. Available from: https://academic.oup.com/bib/article/doi/10.1093/bib/bbae122/7636762 DOI: 10.1093/bib/bbae122
2. Patricia Buendia, Patrick Shironoshita, Lars Eklund, David Molik, Natalie Meyers. System Security Assessment Plan Template. 2025 November. Available from: https://www.rd-alliance.org/wp-content/uploads/2025/11/RDA-AIDV-Template-System-Security-Assessment-Plan.pdf uri: https://www.rd-alliance.org/wp-content/uploads/2025/11/RDA-AIDV-Template-

System-Security-Assessment-Plan.pdf

3. Molik DC, Stahlke AR, Sharma SP, Simmonds TJ, Corpuz RL, Kauwe AN, Schrader JE, Mason CJ, Sim SB, Geib SM. otb: an automated HiC/HiFi pipeline assembles the Prosapia bicincta Genome. G3 (Bethesda). 2024 Aug 7;14(8) PubMed Central PMCID: PMC11304988.
4. Cabugos Leyla, Buble Katheryn, Daenzer Jenna, Jung Sook, Main Dorrie, Marrano Annarita, Molik David, Raciti Daniela, Wright Adam, Yook Karen, Reiser Leonore. AgBioDatabase Finder: an online tool to help researchers find and submit agricultural genomic, genetic, and breeding data. 2025. Available from: https://www.micropublication.org/journals/biology/micropub-biology-001896 DOI: 10.17912/MICROPUB.BIOLOGY.001896
5. Molik DC. met v1: expanding on old estimations of biodiversity from eDNA with a new database framework. Database (Oxford). 2022 May 11;2022 PubMed Central PMCID: PMC9216496.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Cannon E, Molik D, Wright A, Zhang H, Honaas L, Chougule K, Dyer S. Guidelines for gene and genome assembly nomenclature. GENETICS. 2025 March; 229(3):-. Available from: https://academic.oup.com/genetics/article/doi/10.1093/genetics/iyaf006/7954581 DOI: 10.1093/genetics/iyaf006
2. Molik D, Tomlinson D, Davitt S, Morgan E, Sisk M, Roche B, Meyers N, Pfrender M. Combining natural language processing and metabarcoding to reveal pathogen-environment associations. PLOS Neglected Tropical Diseases. 2021; 15(4):e0008755-. Available from: https://dx.plos.org/10.1371/journal.pntd.0008755 DOI: 10.1371/journal.pntd.0008755
3. Molik D, Pfrender M, Emrich S. Uncovering Effects from the Structure of Metabarcode Sequences for Metagenetic and Microbiome Analysis. Methods and Protocols. 2020 March 12; 3(1):22-. Available from: https://www.mdpi.com/2409-9279/3/1/22 DOI: 10.3390/mps3010022
4. Scroggs S, Bird E, Molik D, Nayduch D. Vesicular Stomatitis Virus Elicits Early Transcriptome Response in Culicoides sonorensis Cells. Viruses. 2023 October 18; 15(10):2108-. Available from: https://www.mdpi.com/1999-4915/15/10/2108 DOI: 10.3390/v15102108
5. Drolet B, Reister-Hendricks L, Mayo C, Rodgers C, Molik D, McVey D. Increased Virulence of Culicoides Midge Cell-Derived Bluetongue Virus in IFNAR Mice. Viruses. 2024 September 17; 16(9):1474-. Available from: https://www.mdpi.com/1999-4915/16/9/1474 DOI: 10.3390/v16091474

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Molik, David in SciENcv on 2026-07-27 11:07:30

IDENTIFYING INFORMATION:

NAME: Eggleston, Nicholas

ORCID iD: https://orcid.org/0009-0002-3485-3346

POSITION TITLE: Team Lead/HPC Architect

PRIMARY ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Arkansas, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Western Governors University, Online, Missouri, United States	MS	05/2023 - 10/2024	Information Technology Management
Missouri University of Science and Technology, Rolla, Missouri, USA	BS	08/2011 - 05/2015	Computer Science

Appointments and Positions

2025 - present Team Lead/HPC Architect, Kansas State University, Manhattan, Arkansas, United States

2025 - present Team Lead/HPC Architect, Kansas State University, Manhattan, Kansas, United States

2022 - 2025 Cyberinfrastructure Systems Engineer - Experienced, Federal Reserve Bank of Kansas City, Kansas City, Missouri, United States

2018 - 2022 HPC Engineer, Children's Mercy Hospital, Kansas City, Missouri, United States

2016 - 2018 HPC Administrator, University of Kansas, Lawrence, Kansas, United States

2015 - 2016 HPC Administrator, Missouri University of Science and Technology, Rolla, Missouri, United States

Products

Products Most Closely Related to the Proposed Project

1. Eggleston Nicholas. carcctldr: CaRCC TL;DR. 2026. Available from: https://github.com/nickegg1018/carcctldr uri: https://github.com/nickegg1018/carcctldr
2. Eggleston Nicholas, Smith Matthew. Systems-Facing Track Update. 2026. Available from: https://carcc.org/2026/03/02/2026-carcc-annual-community-update-recap/ uri: https://carcc.org/2026/03/02/2026-carcc-annual-community-update-recap/
3. Eggleston Nicholas. aws-cost-calculator: Amazon Cost Calculator. 2022. Available from: https://github.com/nickegg1018/aws-cost-calculator uri: https://github.com/nickegg1018/aws-cost-calculator
4. Eggleston Nicholas. HPC Administration (Why You Should Do What I Do). 2017.
5. Eggleston Nicholas. Thanatos. 2016. Available from: https://github.com/nickegg1018/thanatos

uri: https://github.com/nickegg1018/thanatos

Other Significant Products, Whether or Not Related to the Proposed Project

1. Eggleston Nicholas. Kickstarting Your Research Computing and Data Career. 2025. Available from: https://carcc.org/2025/05/28/june-community-calls-to-kick-off-summer/ uri: https://carcc.org/2025/05/28/june-community-calls-to-kick-off-summer/

Certification:

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Certified by Eggleston, Nicholas in SciENcv on 2026-07-28 13:51:17

IDENTIFYING INFORMATION:

NAME: Sheldon, Jeffrey

ORCID iD: https://orcid.org/0009-0006-1598-0488

POSITION TITLE: Director

PRIMARY ORGANIZATION AND LOCATION: Kansas State University, K-State Libraries, Sunderland Foundation Innovation Lab, Manhattan, Kansas, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Louisiana State University, Baton Rouge, Louisiana, United States	BFA	08/1993 - 12/2000	Studio Art

Appointments and Positions

2023 - present Director, Kansas State University, K-State Libraries, Sunderland Foundation Innovation Lab, Manhattan, Kansas, United States

2021 - 2023 Associate Director, Kansas State University, K-State Libraries, Sunderland Foundation Innovation Lab, Manhattan, Kansas, United States

2013 - 2021 IT Administrator, Kansas State University, Library Information Technology Services, Manhattan, Kansas, United States

Products

Products Most Closely Related to the Proposed Project

1. Anielak Katy, Whitney Abi. The Role of Students in Academic Makerspaces Panel Discussion. 2025 November. Available from: https://nlc.nebraska.gov/libraries/maker/index.2025.Conference.aspx uri: https://nlc.nebraska.gov/libraries/maker/index.2025.Conference.aspx
2. Harvey Melissa, Su Lishan. Telling Your Science Story: Technology Resources at the Sunderland Foundation Innovation Lab. 2025 March. Available from: https://www.k-state.edu/engagement/interdisciplinary-institutes/water-institute/events/water-wednesdays.html
3. Sheldon Jeffrey. Making the Makerspace: Lessons Learned at the Sunderland Foundation Innovation Lab. 2024 May. Available from: https://www.youtube.com/watch?v=lU9TMniiky4
4. Pierce Stephanie. Engaging Higher Education. 2023 September. Available from: https://www.eventbrite.com/e/2023-maker-space-summit-tickets-635490779027 uri: https://www.eventbrite.com/e/2023-maker-space-summit-tickets-635490779027
5. Rhone Jahvelle, Gates Renee, Sevin Carol, Griffin Shelley, Shippy Kevin. Making the Makerspace: How We Created the Sunderland Foundation Innovation Lab. 2021 May. Available from: https://check2021.ku.edu/session-descriptions#makerspace uri: https://check2021.ku.edu/session-descriptions

Other Significant Products, Whether or Not Related to the Proposed Project

1. Correll Jim, Woods Casey, Towle Nathan. Demystifying Maker Spaces Panel. 2025 September.

Available from: https://emporiamainstreet.com/events/makersummit/ uri: https://emporiamainstreet.com/events/makersummit/

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Sheldon, Jeffrey in SciENcv on 2026-07-28 16:34:17

IDENTIFYING INFORMATION:

NAME: Sibley, Gwendolyn

ORCID iD: https://orcid.org/0000-0001-6552-9472

POSITION TITLE: Scholarly Communication and Data Librarian

PRIMARY ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Emporia State University, Emporia, Kansas, United States	MLS	08/2020 - 05/2022	Librarianship
University of Kansas, Lawrence, Kansas, United States	AB	08/2016 - 05/2020	English Rhetoric, Literature, & Writing

Appointments and Positions

2025 - present Scholarly Communication and Data Librarian, Kansas State University, Manhattan, Kansas, United States

2022 - 2025 Scholarly Communication and Copyright Librarian, Kansas State University, Manhattan, Kansas, United States

Products

Products Most Closely Related to the Proposed Project

1. Spahr, K., McCowen, R., Jackson, C., Otto, R., Sibley, G., Berney, E., Turvey-Welch, M., Misilo, T., Andresen, D., Molik, D. Empowering Data Literacy. K-State Libraries Inaugural Internal Grant; 2024.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Turvey-Welch, M., Sibley, G., Otto, R., Kickbusch, S., Jackson, C., Berney, E. Enjoying "First Sale Doctrine": An Appreciation of library lending from then to now. New Prairie Press; 2025. Available from: https://newprairiepress.org/cgi/viewcontent.cgi?article=1106&context=culsproceedings PMID: https://doi.org/10.4148/ 2160-942X.1106
2. Hall, D., Sibley. G. Welcome to the Legal Circus. US: Kansas Library Association; 2024.
3. Sibley, G. Indigenous Intellectual Property: Growing the Conversation. Association of Tribal Archives, Libraries, & Museums; 2024.
4. Sibley, G. The Incongruous Relationship Between AI Development & University Regulatory Policy. Kraemer Copyright Conference; 2024.
5. Jackson, C., Sibley, G., & Davis, M. Open For All. Open Education Talks; 2024.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Sibley, Gwendolyn in SciENcv on 2026-07-27 17:30:36

Other Personnel Biographical Information

Data Not Available

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Molik, David

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: https://orcid.org/0000-0003-3192-6538

*POSITION TITLE: Department Head, Associate Professor, Librarian

*ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Proposals/Active Projects

***Proposal/Active Project Title:**	RCN: Reimagining a Sustainable Data Network to Accelerate Agricultural Research and Discovery
***Status of Support:**	Current
Proposal/Award Number:	2126334
***Source of Support:**	Phoenix Bioinformatics Corporation, Prime: NSF
***Primary Place of Performance:**	Kansas State University, Manhattan, KS USA
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2021
***Proposal/Active Project End Date: (MM/YYYY):**	08/2026
***Total Anticipated Proposal/Project Amount:**	$95,729

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.75

***Overall Objectives:**

Support coordination, standards development, and community infrastructure for agricultural genomics databases through the AgBioData Research Coordination Network, including activities related to metadata practices, data integration, and community engagement.

***Statement of Potential Overlap:**

No scientific or budgetary overlap. The AgBioData RCN supports community coordination, standards development, and network-building activities for agricultural genomics databases. In contrast, the proposed project focuses on the development and empirical evaluation of machine-actionable publication frameworks and research data infrastructure. There is no duplication of research objectives or committed effort.

***Proposal/Active Project Title:**	CategoryII:K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	National Science Foundation
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	02/2027
***Proposal/Active Project End Date: (MM/YYYY):**	01/2030
***Total Anticipated Proposal/Project Amount:**	$9,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1
2030	0.1

***Overall Objectives:**

Develop and evaluate a federated research service that registers computational environments as citable, version-aware research objects; executes selected workflows through managed research-computing infrastructure; connects computational environment records to repositories, persistent identifiers, and public-access systems; and documents a sustainable and transferable service model for executable and reproducible science.

***Statement of Potential Overlap:**

This is the proposal being submitted. There is no scientific or budgetary duplication with the pending William T. Grant Foundation proposal. K-FORGE develops and evaluates federated infrastructure for executable research objects, whereas the William T. Grant project examines undergraduate publishing outcomes and the use of research evidence in experiential-learning policy. The projects share only David Molik's general expertise in scholarly publishing and research infrastructure and have concurrent effort during 2027–2029. If both projects are funded, effort will be separately allocated at 1.00 person-month per year for K-FORGE and 0.50 person-month per year for the William T. Grant project.

***Proposal/Active Project Title:**	A Research Study of Undergraduate Student Publishing: Impacts on Student Outcomes and Experiential Learning Policies
***Status of Support:**	Pending
Proposal/Award Number:	

***Source of Support:** William T Grant Foundation

***Primary Place of Performance:** Kansas State University, Manhattan, KS, USA

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2027

***Proposal/Active Project End Date: (MM/YYYY):** 12/2029

***Total Anticipated Proposal/Project Amount:** $483,064

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.5
2028	0.5
2029	0.5

***Overall Objectives:**

Examine how undergraduate publishing relates to student outcomes and how evidence about undergraduate publishing is understood and used by higher education decision-makers developing experiential-learning policies. The mixed-methods study will analyze institutional student and publication data, examine disciplinary perspectives on undergraduate publishing, and translate findings into evidence products and policy-engagement activities.

***Statement of Potential Overlap:**

There is no scientific or budgetary duplication with the proposed K-FORGE project. This project studies undergraduate publishing, student outcomes, disciplinary perspectives, and evidence use in higher education policy. K-FORGE develops and evaluates technical infrastructure for registering, executing, preserving, and citing computational research environments. The projects share only David Molik's general expertise in scholarly publishing and research infrastructure and have concurrent effort during 2027–2029. If both projects are funded, the commitments will be separately managed at 0.50 person-month per year for this project and 1.00 person-month per year for K-FORGE.

***Proposal/Active Project Title:** Ag-DEx: A DOI-first Open Data Framework for Agricultural Data Exchange, Discovery, and Governed Reuse

***Status of Support:** Pending

Proposal/Award Number: GRANT14689447, 2026-07893

***Source of Support:** USDA NIFA ODF

***Primary Place of Performance:** Kansas State University, Manhattan, KS USA

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2027

***Proposal/Active Project End Date: (MM/YYYY):** 12/2029

***Total Anticipated Proposal/Project Amount:** $923,325

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.29
2028	0.29
2029	0.29

***Overall Objectives:**

Ag-DEx: Agricultural Data Exchange, Discovery, Governance, and Reuse will develop and evaluate a neutral, producer-centered Open Data Framework for agricultural research and Extension data. The project will establish a trusted repository and stewardship environment; implement persistent identifiers, standardized metadata, provenance, access controls, and OpenAPI-documented interoperability services; and evaluate the framework through TAPS, Agricultural Producer Data Exchange, and EDEN-aligned use cases. The project will also develop governance, stakeholder-engagement, documentation, evaluation, and sustainability models supporting the long-term discovery and responsible reuse of agricultural data.

***Statement of Potential Overlap:**

There is limited conceptual overlap between Ag-DEx and K-FORGE in their use of repository stewardship, metadata, persistent identifiers, provenance, access governance, and sustainability planning. However, the projects have distinct objectives and deliverables. Ag-DEx is a USDA-focused, domain-specific framework for the exchange, governance, publication, and reuse of agricultural research, Extension, and producer-linked data. K-FORGE is a cross-disciplinary research cyberinfrastructure project focused on registering, executing, reproducing, and evaluating computational workflows and associated software environments. There is no duplication of requested costs, equipment, personnel activities, or project deliverables. If both projects are funded, effort and expenses will be managed separately and adjusted as necessary to prevent overcommitment or duplicate charging.

***Proposal/Active Project Title:**	Unlocking Agricultural Research Data for Better Farming Decisions
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	Foundation for Food & Agriculture Research (FFAR)
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2027
***Proposal/Active Project End Date: (MM/YYYY):**	12/2029
***Total Anticipated Proposal/Project Amount:**	$449,912

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1

Year	Person Months
2028	1
2029	1

***Overall Objectives:**

Develop an innovative research program in food and agricultural systems focused on advancing data-driven and computational approaches to improve biological understanding, data integration, and decision-making in agricultural contexts.

***Statement of Potential Overlap:**

No scientific or budgetary overlap. This nomination represents a broad, early-career research program in food and agriculture that is conceptually related to data-intensive biology. The proposed NSF project focuses specifically on the design and evaluation of machine-actionable publication frameworks and research data infrastructure. There is no duplication of specific aims, methods, or committed effort.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

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Certified by Molik, David in SciENcv on 2026-07-27 10:32:43

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Eggleston, Nicholas

*POSITION TITLE: HPC Architect

*ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Proposals/Active Projects

***Proposal/Active Project Title:** Category II: K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Kansas State University, Manhattan, KS, USA

***Proposal/Active Project Start Date: (MM/YYYY):** 02/2027

***Proposal/Active Project End Date: (MM/YYYY):** 01/2030

***Total Anticipated Proposal/Project Amount:** $9,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1
2030	0.1

***Overall Objectives:**

Develop and evaluate a federated research service that registers computational environments as citable, version-aware research objects; executes selected workflows through managed research-computing infrastructure; connects computational environment records to repositories, persistent identifiers, and public-access systems; and documents a sustainable and transferable service model for executable and reproducible science.

***Statement of Potential Overlap:**

This proposal.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Eggleston, Nicholas in SciENcv on 2026-07-27 18:14:52

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Sheldon, Jeffrey

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: https://orcid.org/0009-0006-1598-0488

*POSITION TITLE: Director, Sunderland Foundation Innovation Lab

*ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Proposals/Active Projects

***Proposal/Active Project Title:**	Category II: K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	National Science Foundation
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	02/2027
***Proposal/Active Project End Date: (MM/YYYY):**	01/2030
***Total Anticipated Proposal/Project Amount:**	$9,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1
2030	0.1

***Overall Objectives:**

Develop and evaluate a federated research service that registers computational environments as citable, version-aware research objects; executes selected workflows through managed research-computing infrastructure; connects computational environment records to repositories, persistent identifiers, and public-access systems; and documents a sustainable and transferable service model for executable and reproducible science.

***Statement of Potential Overlap:**

This proposal.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Sheldon, Jeffrey in SciENcv on 2026-07-27 11:16:22

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Sibley, Gwendolyn

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: https://orcid.org/0000-0001-6552-9472

*POSITION TITLE: Scholarly Communication and Data Librarian

*ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Proposals/Active Projects

***Proposal/Active Project Title:**	Category II:K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	National Science Foundation
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	02/2027
***Proposal/Active Project End Date: (MM/YYYY):**	01/2030
***Total Anticipated Proposal/Project Amount:**	$9,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1
2030	0.1

***Overall Objectives:**

Develop and evaluate a federated research service that registers computational environments as citable, version-aware research objects; executes selected workflows through managed research-computing infrastructure; connects computational environment records to repositories, persistent identifiers, and public access systems; and documents a sustainable and transferable service model for executable and reproducible science.

***Statement of Potential Overlap:**

This is the proposal being submitted.There is no scientific or budgetary duplication with the pending William T.

Grant Foundation proposal. Ag-Dex works on building data repository infrastructure. K-FORGE develops and evaluates federated infrastructure for executable research objects, whereas the William T. Grant project examines undergraduate publishing outcomes and the use of research evidence in experiential-learning policy. The projects share only Gwendolyn Sibley's general expertise in scholarly publishing and research infrastructure and have concurrent effort during 2027–2029. If all projects are funded, effort will be separately allocated at .31 person-month for Ag-Dex, 1.00 person-month per year for K-FORGE and 2.00 person-month per year for the William T. Grant project.

***Proposal/Active Project Title:**	A Research Study of Undergraduate Student Publishing: Impacts on Student Outcomes and Experiential Learning Policies
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	William T. Grant
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2027
***Proposal/Active Project End Date: (MM/YYYY):**	12/2029
***Total Anticipated Proposal/Project Amount:**	$483,064

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	2
2028	2
2029	2

***Overall Objectives:**

This research aims to study how undergraduate publishing informs student outcomes and placement, how undergraduate publishing is valued by different disciplines, and how experiential learning policies being implemented at Kansas State University (K-State) and other higher-ed institutions may be informed by undergraduate publishing. This study will analyze institutional student data compared to public publication data while also conducting interviews with faculty on perspectives on undergraduate publishing within disciplines. The methods of this analysis include descriptive statistics, MANOVA, Multiple Linear Regression, Logistic Regression, and Interaction Effects providing a statistically interpretable result. For the qualitative data from the surveys, the analysis will use descriptive statistics, theme coding, and cross tabulation to identify recurring themes and compare perspectives across disciplines, faculty roles, and institutional contexts.

***Statement of Potential Overlap:**

There is no scientific or budgetary duplication with the pending William T. Grant Foundation proposal. Ag-Dex works on building data repository infrastructure. K-FORGE develops and evaluates federated infrastructure for executable research objects, whereas the William T. Grant project examines undergraduate publishing outcomes and the use of research evidence in experiential-learning policy. The projects share only Gwendolyn Sibley's general expertise in scholarly publishing and research infrastructure and have concurrent effort during

2027–2029. If all projects are funded, effort will be separately allocated at .31 person-month for Ag-Dex, 1.00 person-month per year for K-FORGE and 2.00 person-month per year for the William T. Grant project.

***Proposal/Active Project Title:**	Ag-DEx: A DOI-first Open Data Framework for Agricultural Data Exchange, Discovery, and Governed Reuse
***Status of Support:**	Pending
Proposal/Award Number:	GRANT14689447, 2026-07893
***Source of Support:**	USDA NIFA
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2027
***Proposal/Active Project End Date: (MM/YYYY):**	12/2029
***Total Anticipated Proposal/Project Amount:**	$923,325

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.31
2028	0.31
2029	0.31

***Overall Objectives:**

Ag-DEx will create a neutral, secure, and sustainable open data framework for agricultural research, Extension, and producer-centered data. The project will deploy an InvenioRDM-based repository backend, persistent identifier workflows, metadata templates, access categories, OpenAPI documented compatibility interfaces, and cyberinfrastructure connections for storage, transfer, backup, and recovery. Ag-DEx will demonstrate and evaluate the framework through three applied use cases: a TAPS data hub for multi-state agricultural research and Extension data; an Ag Producer Data Exchange workflow for producer-controlled data packages; and an Extension/Natural Disasters and NABC-supported EDEN-aligned documentation workflow for emergency-scaling and practical Extension use. The project will also establish governance rules, stakeholder advising, and a sustainability model to make agricultural data easier to discover, cite, publish, and reuse while maintaining producer control and project boundaries.

***Statement of Potential Overlap:**

There is no scientific or budgetary duplication with the pending William T. Grant Foundation proposal. Ag-Dex works on building data repository infrastructure. K-FORGE develops and evaluates federated infrastructure for executable research objects, whereas the William T. Grant project examines undergraduate publishing outcomes and the use of research evidence in experiential-learning policy. The projects share only Gwendolyn Sibley's general expertise in scholarly publishing and research infrastructure and have concurrent effort during 2027–2029. If all projects are funded, effort will be separately allocated at .31 person-month for Ag-Dex, 1.00 person-month per year for K-FORGE and 2.00 person-month per year for the William T. Grant project.
The Data Librarian will support metadata, curation, dataset citation, documentation, repository policy alignment, training materials, and FAIR/TRUST-aligned stewardship workflows.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Sibley, Gwendolyn in SciENcv on 2026-07-27 17:29:21

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Molik, David	Kansas State University	

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Zimmer, Andrew	Family	amzimmer2@wisc.edu	
R	Zimmer, Margaret	Family	margaret.zimmer@wisc.edu	
R	Roche, Benjamin	Personal	benjamin.roche@UND.edu	
R	Coggins, Bret	Personal	bcoggins@nd.edu	
R	Molla Morales, Almudena	Personal	Gregor Mendel Institute	

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	Pfrender, Michael	University of Notre Dame	Biology
T	Archie, Elizabeth	University of Notre Dame	Biology
T	Meyers, Natalie	CNI-ARL	
T	Perkins, Alex	University of Notre Dame	Biology
T	Emrich, Scott	University of Tennessee Knoxville	CSE

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
A	Stacey L. P. Scroggs	Kansas State University / USDA ARS		
A	Edward J. Bird	USDA Agricultural Research Service		
A	Dana Nayduch	USDA Agricultural Research Service		
A	Adam J. Wright	Ontario Institute for Cancer Research (OICR)		
A	Mark D. Wilkinson	Universidad Politecnica de Madrid		
A	Christopher J. Mungall	Lawrence Berkeley National Laboratory		
A	Scott Cain	USDA Agricultural Research Service		
A	Stephen Richards	Baylor College of Medicine		
A	Paul W. Sternberg	California Institute of Technology		
A	Ellen Provin	Texas A&M University		
A	Jonathan L. Jacobs	Lawrence Berkeley National Laboratory		
A	Scott M. Geib	USDA Agricultural Research Service		

A	Daniela Raciti	USDA Agricultural Research Service		
A	Karen J. Yook	California Institute of Technology		
A	Lincoln Stein	Ontario Institute for Cancer Research		
A	Amanda R. Stahlke	USDA Agricultural Research Service		
A	Sharu P. Sharma	Iowa State University		
A	Tyler J. Simmonds	USDA Agricultural Research Service		
A	Renee L. Corpuz	USDA Agricultural Research Service		
A	Angela N. Kauwe	USDA Agricultural Research Service		
A	Jeremy E. Schrader	USDA Agricultural Research Service		
A	Charles J. Mason	USDA Agricultural Research Service		
A	Sheina B. Sim	USDA Agricultural Research Service		
A	Katheryn Buble	USDA Agricultural Research Service		
A	Leyla Cabugos	USDA Agricultural Research Service		
A	Jaqueline Campbell	USDA Agricultural Research Service		
A	Jenna Daenzer	USDA Agricultural Research Service		
A	Leonore Reiser	The Arabidopsis Information Resource (TAIR)		
A	Sook Jung	Washington State University		
A	Joaquin Carbonara	Buffalo State University		
A	Halfon, Marc S.	University at Buffalo-State University of New York		07/23/26
A	Olga S. Novikova	Buffalo State University		
A	Jimmy Pitzer	USDA Agricultural Research Service		
A	Brian E. Scheffler	USDA Agricultural Research Service		
A	Phillip Shults	USDA Agricultural Research Service		
A	Caitlin Taylor	USDA Agricultural Research Service		
A	Ethalinda K. S. Cannon	USDA Agricultural Research Service		
A	Huiting Zhang	USDA Agricultural Research Service		
A	Loren Honaas	USDA Agricultural Research Service		
A	Kapeel Chougule	USDA Agricultural Research Service		
A	Sarah Dyer	USDA Agricultural Research Service		
A	Tony Boston	Research Data Alliance		
A	Massamba Diop	Research Data Alliance		
A	Debora Drucker	Research Data Alliance		
A	Matt Fry	Research Data Alliance		
A	Sylvain Grellet	Research Data Alliance		
A	Stefanie Kethers	University of Sydney		
A	Viorica Lupu	Research Data Alliance		
A	Valeria Pesce	Research Data Alliance		

A	Victoria Pickens	USDA Agricultural Research Service		
A	Cassandra Olds	USDA Agricultural Research Service		
A	Kristopher Silver	USDA Agricultural Research Service / Kansas State University		
A	Dorrie Main	Washington State University		
A	Annarita Marrano	Washington State University		
C	Monica Poelchau	USDA Agricultural Research Service		
C	Sue Brown	Kansas State University		
C	Brad Coates	USDA Agricultural Research Service		
C	Marc Halfon	University at Buffalo (SUNY)		
C	Duane McKenna	University of Memphis		
C	Kevin Hackett	USDA Agricultural Research Service		
C	Carson Andorf	USDA Agricultural Research Service		
C	Sarah Dyer	EMBL-EBI		
C	Sunita Kumari	Cold Spring Harbor Laboratory		
C	Dorrie Main	Washington State University		
C	John McNamara	Washington State University		
C	Sushma Naithani	Oregon State University		
C	Monica Poelchau	USDA Agricultural Research Service		
C	Leonore Reiser	Phoenix Bioinformatics		
C	Peter Selby	Cornell University		
C	Taner Sen	USDA Agricultural Research Service		
C	Marcela Karey Tello-Ruiz	Phoenix Bioinformatics		
C	Ryan Otto	Kansas State University Libraries		
C	Gwendolyn Sibley	Kansas State University Libraries		
C	Carolyn Jackson	Kansas State University Libraries		
A	Tvedte, Eric S.	National Institutes of Health / NLM / NCBI		07/23/26
A	Bucher, Gregor	University of Göttingen		07/23/26
A	Luecke, David M.	USDA Agricultural Research Service		07/23/26
A	Sylvester, Terrence	University of Memphis		07/23/26
A	Blaxter, Mark	Wellcome Sanger Institute		07/23/26
A	Elsik, Christine G.	University of Missouri		07/23/26
A	Howe, Kerstin	Wellcome Sanger Institute		07/23/26
A	McKenna, Duane D.	University of Memphis		07/23/26
A	Murphy, Terence D.	National Institutes of Health / NLM / NCBI		07/23/26
A	Schrader, Lukas	University of Münster		07/23/26
A	Sotero-Caio, Cibele G.	Wellcome Sanger Institute		07/23/26

A	Waterhouse, Robert M.	SIB Swiss Institute of Bioinformatics		07/23/26
A	Childers, Anna K.	USDA Agricultural Research Service		07/23/26

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date
B	Bowman-Zatzkin, Whitney	BIO-ISAC	Cyberbiosecurity Quarterly	
E	Guise Nicholas	Georgia Tech Research Institue	Cyberbiosecurity Quarterly	
E	Mayo, Keith	TIIAS	Cyberbiosecurity Quarterly	
E	Reid, Bill	Google Cloud	Cyberbiosecurity Quarterly	
E	Self, Adrian	Kansas State University	Cyberbiosecurity Quarterly	
E	Vasserman, Eugine	Kansas State University	Cyberbiosecurity Quarterly	
B	Puzis, Rami	Ben-Gurion University of the Negev	IEEE cyberbio26 Program Committee	
E	Beal, Jake	RTX	IEEE cyberbio26 Program Committee	
E	Douglas, Gavin	University of New Brunswick	PCI Genomics	

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Eggleston, Nicholas J	Kansas State University	

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Andresen, Daniel	Professional	dan@ksu.edu	
R	Turner, David	Professional	daveturner@ksu.edu	
R	Wells, Nathan	Professional	nathanrwells@ksu.edu	

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
C	Molik, David	Kansas State University	molik@ksu.edu	
C	Sheldon, Jeffrey	Kansas State University	jsheldon@ksu.edu	
C	Sibley, Gwendolyn	Kansas State University	grsibley@ksu.edu	
C	Lougee, BJ	Pacific Northwest National Laboratory		12/31/25
C	Woodward, Greg	Federal Reserve Bank of Kansas City		12/31/25
C	Yang, Shu-Kuei	Federal Reserve Bank of Kansas City		12/31/25
C	Dice, Jacob	Federal Reserve Bank of Kansas City		12/31/25

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Jeffrey Sheldon	Kansas State University	

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Anna Zemlyanova	Spouse	azemlyanova@ksu.edu Mathematics	

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	None		
T	None		

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
A	Stephanie Pierce	Innovate24	stephanie@innovate24.org	09/20/23
A	Melissa Harvey	Kansas Water Institute, Kansas Water Institute	mharvey@ksu.edu	03/11/26
A	Lishan Su	Kansas State University, Kansas Water Institute	lishans@ksu.edu	03/11/26
A	Jahvelle Rhone			06/15/26
C	Renee Gates	Kansas State University, Sunderland Foundation Innovation Lab	regates@ksu.edu	
C	Kevin Shippy	Kansas State University, Sunderland Foundation Innovation Lab	kshippy@ksu.edu	
C	Shelley Griffin	Kansas State University, Sunderland Foundation Innovation Lab	shegriff@ksu.edu	
C	Carol Sevin	Kansas State University, Sunderland Foundation Innovation Lab	sevin@ksu.edu	
C	Cheryl Bachelor Boothe	Kansas State University, Sunderland Foundation Innovation Lab	cbachelorboothe@ksu.edu	
C	Charles Appelseth	Kansas State University, Sunderland Foundation Innovation Lab	chasapp@ksu.edu	
C	Jana Thomas	Kansas State University, A Q Miller School of Media and Communication	janamthomas@ksu.edu	
C	Russ Feldhausen	Kansas State University, Computer Science	russfeld@ksu.edu	
C	Abi Whitney	Johnson County Library		11/05/25
C	Katy Anielak	Creighton Univeristy, Libraries	katyanielak@creighton.edu	11/05/25

C	David Molik	Kansas State University, Libraries	molik@ksu.edu	
C	Gwendolyn Sibley	Kansas State University, Libraries	grsibley@ksu.edu	
C	Nicholas Eggleston	Kansas State University, Computer Science	njeggleston@ksu.edu	

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date
B	None			
E	None			

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Gwendolyn Sibley	Kansas State University	
		University of Kansas (collaborator)	07/24/26

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Phelps, Zane	Family		

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	none		
T	none		

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
C	Anderson, Alice	Kansas State University		
A	Andresen, Dan	Kansas State University		
C	Berney, Elizabeth			
C	Brooks-Kieffer, Jamene	University of Kansas		
C	Coleman, Jason	Kansas State University		
A	Crawley, Francis			
A	Davis, Makay			
C	Deiser, Raymond	Kansas State University		
C	Durham, E	University of Kansas		
A	Eggleston, Nicholas	Kansas State University		
C	Glymour, Bruce	Kansas State University		
A	Hale, Raelynne	Kansas State University		
A	Hall, Danielle	Kansas State University		
C	Hitzler, Pascal	Kansas State University		
A	Jackson, Carolyn	Kansas State University		
A	Kickbusch, Scott	Kansas State University		
A	McCowen, Robert			
A	Meyers, Natalie	UC San Diego		
A	Misilo, Tom	Kansas State University		

A	Mocnik, Joe	Kansas State University		
A	Molik, David	Kansas State University		
C	Olds, Cassandra	Kansas State University		
C	Olsen, Livia	Kansas State University		
A	Otto, Ryan	Kansas State University		
A	Peck, Alex			
A	Schultz, Tristan (Nova)			
C	Sevin, Carol	Kansas State University		
A	Sheldon, Jeffrey	Kansas State University		
C	Shippy, Teresa	Kansas State University		
C	Slimp, Mickey	Great Plains Network		
C	Spahr, Kendra	Kansas State University		
C	Tank, William	Kansas State University		
A	Turvey-Welch, Michelle	Kansas State University		
C	Vasserman, Eugene	Kansas State University		
C	Van Ittersum, Christina	Kansas State University		
A	Wilkinson, Mark			
A	Wiser, James	Kansas State University		
A	Wright, Adam			

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date
B	none			
E	none			

Name: Molik, David C.
ORCID iD: https://orcid.org/0000-0003-3192-6538
Position Title: Department Head, Associate Professor, Librarian
Primary Organization and Location: Kansas State University, Manhattan, Kansas, United States

Synergistic Activities:

1. Leads development of Kansas State University's campus research data repository through the Open Publishing Exchange. The InvenioRDM-based service supports DataCite DOI minting, ORCID integration, institutional authentication, preservation-ready storage, interoperable metadata, and researcher guidance, creating a campus pathway for publishing, preserving, citing, and responsibly sharing research data.
2. Develops and contributes to open-source, containerized computational workflows and research software for genomics and metagenomics, including automated genome-assembly and antimicrobial-resistance pipelines. These tools make complex analyses more reproducible, portable, and reusable across institutional research-computing environments.
3. Contributes to national and international genomic data standards, including machine-actionable metadata and nomenclature conventions that connect genome assemblies, provenance, software environments, and repository records. This work translates FAIR and TRUST principles into implementable practices for databases, publishers, and researchers.
4. Serves on the AgBioData Steering Committee for the NSF Research Coordination Network, coordinating agricultural genomics databases around shared standards, cross-database discovery, training, governance, and sustainability. This service transfers infrastructure and data-management practices among distributed community resources and their users.
5. Chairs the i5k Training Working Group, developing documentation, workshops, and training that help researchers and students analyze, document, and publish insect genome data using reproducible practices. The activity broadens access to genome-informatics expertise across institutions and career stages.

Synergistic Activities: Sibley, Gwendolyn
ORCID iD: 0000-0001-6552-9472
Position Title: Scholarly Communication and Data Librarian, Assistant Professor
Primary Organization and Location: Kansas State University, Manhattan, KS, United States

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1. Main lead of K-State campus research-data services which support faculty, staff, and students who have research data needs. Services include all aspects of the data life cycle including, generation, collection, processing, storage, interpretation, visualization, analysis, and management.
2. Provides reference for repository and data-management support through expertise of various repository services, including Dryad, Zenodo, and other specialist repositories. Also communicates guidance of data management plan systems, particularly through national grant institutions, such as the NIH and the NSF.
3. Give guidance on FAIR, CARE and AI-ready Data Practices, which are standards that endeavor to help researchers make data more discoverable, reusable, ethically sourced, and connected to publications.
4. Facilitate scholarly communication and data-literacy training by educating researchers on their author rights as they look to publish their data and corresponding papers of their research findings. This ranges from educating about adequate citation and credit to helping understand legal language in publishing contracts.
5. Educate students and researchers on standards and professional-community service, particularly within the academic publishing landscape as opposed to the private or governmental sectors of information sharing practices.

Name: Sheldon, Jeffrey, Thomas
ORCID iD: https://orcid.org/0009-0006-1598-0488
Position Title: Director, Sunderland Foundation Innovation Lab
Primary Organization and Location: Kansas State University, Manhattan, Kansas, 66506

Synergistic Activities:

1. Leadership of the Sunderland Foundation Innovation Lab - Direct the Sunderland Foundation Innovation Lab at Kansas State University Libraries, expanding access to emerging technologies, digital fabrication, artificial intelligence, and experiential learning opportunities for students, faculty, researchers, and community partners.
2. Advancing Digital Innovation Through Cross-Sector Engagement - Serve as Co-Chair of the K-State Digital Summit, bringing together higher education, government, military, industry, and community stakeholders to foster collaboration around digital innovation, workforce development, and emerging technologies. The initiative was recognized through K-State's One K-State Excellence in Engagement Award.
3. Dissemination of Makerspace and Innovation Practices - Share lessons learned in makerspace development, innovation programming, and technology-enabled learning through presentations and panel discussions at venues including the CHECK Conference, Maker Space Summit, and Nebraska Makerspace Conference.
4. Support for Interdisciplinary Research and Project Development - Facilitate connections between researchers and emerging technologies through workshops, demonstrations, and collaborative programming, including activities with the Kansas Water Institute and participation in interdisciplinary initiatives associated with the Institute for Digital Agriculture and Advanced Analytics (ID3A).
5. Student Workforce Development and Experiential Learning - Develop opportunities for students to gain hands-on experience with emerging technologies, prototyping tools, makerspace resources, and innovation-focused projects that connect academic learning with practical applications and community engagement.

Name: Sheldon, Jeffrey, Thomas
ORCID iD: https://orcid.org/0009-0006-1598-0488
Position Title: Director, Sunderland Foundation Innovation Lab
Primary Organization and Location: Kansas State University, Manhattan, Kansas, 66506

Synergistic Activities:

1. Leadership of the Sunderland Foundation Innovation Lab - Direct the Sunderland Foundation Innovation Lab at Kansas State University Libraries, expanding access to emerging technologies, digital fabrication, artificial intelligence, and experiential learning opportunities for students, faculty, researchers, and community partners.
2. Advancing Digital Innovation Through Cross-Sector Engagement - Serve as Co-Chair of the K-State Digital Summit, bringing together higher education, government, military, industry, and community stakeholders to foster collaboration around digital innovation, workforce development, and emerging technologies. The initiative was recognized through K-State's One K-State Excellence in Engagement Award.
3. Dissemination of Makerspace and Innovation Practices - Share lessons learned in makerspace development, innovation programming, and technology-enabled learning through presentations and panel discussions at venues including the CHECK Conference, Maker Space Summit, and Nebraska Makerspace Conference.
4. Support for Interdisciplinary Research and Project Development - Facilitate connections between researchers and emerging technologies through workshops, demonstrations, and collaborative programming, including activities with the Kansas Water Institute and participation in interdisciplinary initiatives associated with the Institute for Digital Agriculture and Advanced Analytics (ID3A).
5. Student Workforce Development and Experiential Learning - Develop opportunities for students to gain hands-on experience with emerging technologies, prototyping tools, makerspace resources, and innovation-focused projects that connect academic learning with practical applications and community engagement.

Data Management and Sharing Plan

Submitted to:
Direct For Computer & Info Scie & Enginr (CSE), Office of Advanced Cyberinfrastructure (OAC) (OAC), Data Cyberinfrastructure

1. Software, APIs, and Technical Documentation

Data or Research Product Category General Information

Data or Research Product Category: **Software, APIs, and Technical Documentation**

Description: **Source code and software releases for the K-FORGE registry, APIs, workflow-execution services, user interfaces, deployment and infrastructure-as-code configurations, test suites, schemas, and technical and user documentation. Credentials, private keys, vulnerability details, and security-sensitive production configurations will not be included in public releases.**

Access Policies and Limitations

Access Limitation(s)/Exception(s):

1. **Security or Cybersecurity Restrictions**

Explanation(s) and Justification(s) of Concern(s): **Public releases will omit passwords, credentials, cryptographic keys, unpatched vulnerability information, sensitive network topology, and production-only security configurations. These exclusions are necessary to protect K-FORGE infrastructure and users. All non-sensitive source code, schemas, documentation, and release artifacts will be shared publicly.**

Data Standards and Metadata

Data/Metadata Standard: **Semantic Versioning 2.0.0**

Additional Details: **Software and technical products will use Git version-control metadata, Semantic Versioning 2.0.0, tagged releases, commit identifiers, and checksums. Container artifacts will follow applicable Open Container Initiative specifications, and APIs will be documented using the OpenAPI Specification. Citation and dependency metadata will use Citation File Format, SPDX, and/or CycloneDX. Configuration and machine-readable documentation will use JSON, YAML, Markdown, and other openly documented formats as appropriate.**

Data or Research Product Provenance

Data Source: **New Data Collection**

Additional Details: **Provenance will be documented through repository histories, commit identifiers, release tags, continuous-integration records, build manifests, software bills of materials, checksums, and links between software versions and associated research products.**

Public Archiving

Data Sharing Location: **https://krex.k-state.edu/home**

Additional Details: **Active development will occur in a publicly accessible version-controlled repository. Major tagged releases and accompanying documentation will be archived in a persistent repository such as K-State Research Exchange (K-REx), with persistent identifiers assigned where appropriate. K-REx is expected to support DOI assignment by the start of the project; if DOI assignment is not yet available, applicable products will be archived in another recognized repository that assigns persistent identifiers.**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Additional Details: **Software will be released on a rolling basis following quality, licensing, and security review. Versions supporting publications will be publicly available at or before publication. Major archival releases will occur at project milestones, with final project releases available no later than the end of the award.**

Data Availability

Data Availability Period: **At least 10 years**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Nicholas Eggleston**

Role: **co-Principal Investigator**

Organization Associated within the Proposal: **Kansas State University**

Email Address: **njeggleston@ksu.edu**

2. Executable Research Objects and Registry Metadata

Data or Research Product Category General Information

Data or Research Product Category: **Executable Research Objects and Registry Metadata**

Description: **Machine-readable metadata and research objects produced or curated during K-FORGE development and validation, including workflow definitions, container references, notebooks, software-environment descriptors, model descriptors, provenance records, PID and DOI metadata, citation files, example datasets, and reproducibility packages. External objects will retain their original source, license, identifier, and provenance information.**

Access Policies and Limitations

Not applicable

Data Standards and Metadata

Data/Metadata Standard: **RO-Crate 1.2**

Additional Details: **Metadata will use appropriate open and machine-readable formats, including JSON, JSON-LD, DataCite metadata, schema.org terms, ORCID identifiers, DOI and other persistent identifiers, RO-Crate or equivalent research-object packaging, W3C PROV-compatible provenance, OCI descriptors, checksums, citation metadata, and native workflow formats where applicable**

Data or Research Product Provenance

Data Source: **New Data Collection and Existing Data or Data Reuse**

Additional Details: **K-FORGE-generated records will identify the creating software version, responsible agent, creation date, source object, workflow, inputs, outputs, licenses, identifiers, and checksums. Reused or externally contributed artifacts will retain links to their original repositories, identifiers, versions, creators, and license terms.**

Public Archiving

Data Sharing Location: **https://krex.k-state.edu/home**

Additional Details: **Public records and research objects will be available through the K-FORGE registry. Preservation copies of significant releases, schemas, metadata exports, and demonstration research objects will also be deposited in K-REx or another recognized persistent repository, with DOI or comparable PID assignment where appropriate.**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Additional Details: **Public research objects and metadata will normally be available upon registration or release. Products supporting publications will be available at or before publication. Final schemas, metadata exports, and demonstration packages will be archived no later than the end of the award.**

Data Availability

Data Availability Period: **At least 10 years**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **David Molik**

Role: **Principal Investigator**

Organization Associated within the Proposal: **Kansas State University**

Email Address: **molik@ksu.edu**

3. Benchmarking, Validation, and Operational Data

Data or Research Product Category General Information

Data or Research Product Category: **Benchmarking, Validation, and Operational Data**

Description: **Data generated through performance testing, reproducibility validation, early operations, and full operations, including benchmark results, workflow-execution outcomes, reliability metrics, aggregate usage statistics, uptime and error summaries, and system-evaluation measures. Public datasets will exclude credentials, identifiable user information, sensitive network details, and information that would create a cybersecurity risk.**

Access Policies and Limitations

Access Limitation(s)/Exception(s):

1. **Security or Cybersecurity Restrictions**

Explanation(s) and Justification(s) of Concern(s): **Public releases will contain sanitized or aggregated benchmark, validation, reliability, and usage data. Raw logs containing user identifiers, IP addresses, credentials, sensitive network information, vulnerability information, or other operational-security details will not be released. Restrictions are necessary to protect users and the security and integrity of the national-scale service.**

Data Standards and Metadata

Data/Metadata Standard: **Frictionless Data Package Specification**

Additional Details: **Curated benchmark and validation datasets will use the Frictionless Data Package Specification where applicable, with nonproprietary formats such as CSV, JSON, JSON-LD, or Parquet. Records will include documented schemas, data dictionaries, units, ISO 8601 dates and timestamps, software and hardware versions, benchmark definitions, and machine-readable provenance. Results will be linked to the applicable software release and test configuration.**

Data or Research Product Provenance

Data Source: **New Data Collection**

Additional Details: **Data will be generated by K-FORGE test suites, workflow executions, infrastructure monitoring, validation procedures, and project evaluation activities. Records will identify the software version, hardware or execution environment, test configuration, date, responsible component, input artifact, and applicable workflow or benchmark.**

Public Archiving

Data Sharing Location: **https://krex.k-state.edu/home**

Additional Details: **Curated and sanitized benchmark and validation datasets will be deposited in K-REx, or another appropriate persistent repository. Public summary metrics may also be displayed through K-FORGE, but the operational service will not serve as the sole preservation location.**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Additional Details: **Data supporting publications will be released at or before publication. Other curated benchmark and validation datasets will be released following validation and security review, ordinarily within 12 months of the relevant project phase and no later than the end of the award.**

Data Availability

Data Availability Period: **At least 10 years**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Nicholas Eggleston**

Role: **co-Principal Investigator**

Organization Associated within the Proposal: **Kansas State University**

Email Address: **njeggleston@ksu.edu**

4. Training, Outreach, and Evaluation Products

Data or Research Product Category General Information

Data or Research Product Category: **Training, Outreach, and Evaluation Products**

Description: **Workshop curricula, tutorials, documentation, training exercises, demonstration datasets, survey instruments, de-identified or aggregated participant feedback, evaluation datasets, and evaluation reports produced through K-FORGE education, outreach, and independent project-evaluation activities. Identifiable participant information will be protected in accordance with applicable institutional review, consent, and privacy requirements.**

Access Policies and Limitations

Access Limitation(s)/Exception(s):

1. **Human Data Protection**
2. **Security or Cybersecurity Restrictions**

Explanation(s) and Justification(s) of Concern(s): **Curricula, tutorials, instruments, and aggregate evaluation findings will be publicly shared. Direct identifiers and combinations of information that could reasonably**

identify workshop participants, interviewees, survey respondents, or staff will not be publicly released. Any individual-level data will be de-identified and shared only when consistent with consent, applicable institutional review determinations, and K-State policies

Data Standards and Metadata

Data/Metadata Standard: **Web Content Accessibility Guidelines (WCAG) 2.2**

Additional Details: **Public-facing training and outreach products will conform to the Web Content Accessibility Guidelines (WCAG) 2.2 to the extent applicable. Materials will use accessible and reusable formats such as HTML, Markdown, PDF/A, CSV, JSON, and open document formats. Evaluation datasets will include codebooks, variable definitions, instruments, analysis documentation, and applicable metadata. Accessibility, licensing, and version information will accompany publicly released educational materials.**

Data or Research Product Provenance

Data Source: **New Data Collection**

Additional Details: **Products will be generated through K-FORGE workshops, training activities, community engagement, participant feedback, and independent project evaluation. Versions of instruments, curricula, analysis code, and reports will be documented, and de-identification or aggregation procedures will be recorded.**

Public Archiving

Data Sharing Location: **https://krex.k-state.edu/home**

Additional Details: **Public training materials, instruments, de-identified datasets, and evaluation reports will be preserved in K-REx, or another recognized persistent repository. Restricted identifiable records, when collected, will remain in approved K-State systems and will not be deposited publicly.**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Additional Details: **Training materials will be released when used or shortly thereafter. De-identified evaluation products will be released with the associated report or publication after privacy review. Final curricula, instruments, and evaluation reports will be publicly archived no later than the end of the award.**

Data Availability

Data Availability Period: **At least 10 years**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Jeffrey Sheldon**

Role: **co-Principal Investigator**

Organization Associated within the Proposal: **Kansas State University**

Email Address: **jsheldon@ksu.edu**

MENTORING PLAN

Program structure. K-FORGE will support two graduate researchers employed through the Open Publishing Exchange at Kansas State University Libraries. They will form the initial cohort of the Libraries' Digital Scholarship Training Program, a paid, embedded research-training program combining substantive project responsibilities with technical instruction, evidence-based pedagogy, professional development, and career preparation. The model adapts a comparable digital-scholarship graduate program previously experienced by PI David Molik. Students will retain the disciplinary mentoring and degree requirements of their home departments while receiving dedicated workspace in Hale Library, integration into the OPE and K-FORGE teams, and a monthly Digital Scholarship Seminar and community-of-practice meeting.

Individualized mentoring and supervision. Molik will serve as primary supervisor and will meet with the students weekly, individually or as a cohort, to review progress, address challenges, and connect assignments to academic and career goals. Each student will complete an Individual Development Plan identifying technical competencies, research products, professional-development goals, and career pathways. Progress will be reviewed quarterly and the plan updated annually. Gwendolyn Sibley will provide additional mentoring in research data management, metadata, curation, documentation, citation, repository practice, and FAIR- and TRUST-aligned stewardship.

Research, technical, and instructional training. Students will receive hands-on training in workflow registration, persistent identifiers, metadata standards, research-object packaging, provenance, software and container documentation, repository deposit, reproducibility assessment, and public dissemination. Assignments may include preparing metadata and workflow records, evaluating research objects, documenting pilots, supporting repository integration, analyzing validation results, developing training materials, and synthesizing stakeholder feedback. Instructional development will progress from observing experienced instructors, to co-designing and co-teaching a workshop, to delivering a supervised session with structured feedback. Jeffrey Sheldon and Sibley will provide opportunities to practice instruction with learners from varied disciplinary and technical backgrounds.

Interdisciplinary and collaborative development. Students will participate in project meetings and work with personnel from OPE, Beocat, the Sunderland Foundation Innovation Lab, and project evaluation and advisory activities. Nicholas Eggleston and Beocat personnel will provide exposure to managed execution, research computing, containers, monitoring, reliability, and cybersecurity. Sheldon and Sunderland personnel will provide experience in workflow prototyping, heterogeneous and edge-AI environments, instructional design, workshops, and learner support. These interactions will develop collaboration across disciplinary, organizational, and professional boundaries.

Responsible and ethical professional practice. Mentoring will include required university training and project-specific guidance in responsible and ethical conduct of research, research security, cybersecurity, privacy, authorship, attribution, licensing, accessibility, data stewardship, reproducibility, version control, documentation, and appropriate handling of user and operational information. Students will maintain transparent records of methods and contributions and follow applicable institutional and project policies.

Communication, leadership, and career preparation. Students will contribute to professional products such as documentation, software or metadata releases, training materials, posters, presentations, evaluation reports, manuscripts, and proposals. Mentors will provide feedback on technical writing, presentations, teaching, facilitation, project management, teamwork, and mentoring others. Career counseling will address opportunities in research cyberinfrastructure, libraries, scholarly communication, data stewardship, government, higher education, nonprofit organizations, and technology-oriented research services.

Assessment. Mentoring effectiveness will be assessed through weekly feedback, quarterly milestone reviews, annual Individual Development Plan updates, completed technical, instructional, and scholarly products, student self-assessment of competencies, and opportunities for direct or confidential feedback. Mentoring activities and outcomes will be documented for annual and final NSF reporting.

KANSAS STATE UNIVERSITY | Carl R. Ice College of Engineering
Department of Computer Science

July 22, 2026

To: NSF Program Officer

If the proposal submitted by Dr. David Molik entitled "K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science," is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment and Other Resources section of the proposal.

Sincerely,



Daniel Andresen, Ph.D.
Professor and Department Head
Department of Computer Science
Director, Institute for Computational Research in Engineering and Science

2184 Engineering Hall, Manhattan, KS 66506-2302 | (785) 532-6350 | fax (785) 532-7353 | csoffice@ksu.edu | cs.ksu.edu

KANSAS STATE UNIVERSITY | Office of Educational Innovation and Evaluation

July 8, 2026

National Science Foundation
Randolph Building
401 Dulany Street
Alexandria, VA 22314

To whom it may concern:

If the proposal submitted by Dr. David Molik entitled "K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science" is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal.

Sincerely,



LeAnn Brosius, M.A.
Executive Director, Office of Educational Innovation & Evaluation

1114 Mid-Campus Drive North, 004 Bluemont Hall, Manhattan, Kansas 66506 | 785-532-5930 | Fax: 785-532-7815 | oeie@ksu.edu

KANSAS STATE UNIVERSITY

July 15, 2026

Re: Letter of Collaboration for K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science

Dear Review Panel:

If the proposal submitted by Dr. David Molik entitled "K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science" is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment, and Other Resources section of the proposal.

Sincerely,



Susan Metzger

Director, Strategic Interdisciplinary Program Development
Director, Kansas Water Institute (KWI)
Director, Institute for Digital Agriculture and Advanced Analytics (ID3A) – Engagement
Director, Kansas Center for Agricultural Resources and the Environment (KCARE)
Office of the Vice President for Research

GREAT PLAINS NETWORK

CONNECTING NETWORKS • ADVOCATING RESEARCH • CULTIVATING COMMUNITY

July 22, 2026

National Science Foundation
2415 Eisenhower Avenue
Alexandria, VA 22314

Dear NSF Program Officer:

If the proposal submitted by Dr. David Molik entitled 'K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science' is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal.

Sincerely,



Mickey Slimp, Ed.D., M.P.H., M.A.
Executive Director

KANSAS STATE UNIVERSITY

NABC | National Agricultural Biosecurity Center
KANSAS STATE UNIVERSITY

1041 Pat Roberts Hall, 1900 Denison Ave., Manhattan, KS 66506-7600 | 785-532-6193 | nabc@bri.ksu.edu

July 17, 2026

To Whom It May Concern,

If the proposal submitted by Dr. David Molik entitled 'K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science' is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal.
Sincerely,

Marty Vanier DVM

Marty Vanier, DVM
Director, National Agricultural Biosecurity Center
1041 Pat Roberts Hall
Kansas State University
Manhattan, KS 66506

Desk: 785.532.3929
Cell: 785.236.0830

mvanier@bri.ksu.edu

KANSAS STATE UNIVERSITY

Jane M. Schuh, Ph.D.
Director | Kansas Agricultural Experiment Station
Associate Dean of Research and Graduate Programs | College of Agriculture

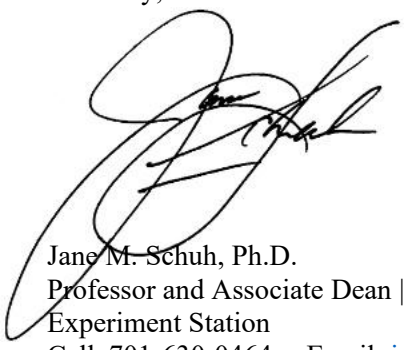
July 22, 2026

Re: Letter of Collaboration — Category II: K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science

Dear NSF Program Officer:

If the proposal submitted by Dr. David C. Molik entitled "Category II: K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science" is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal.

Sincerely,



Jane M. Schuh, Ph.D.
Professor and Associate Dean | Ag Research and Graduate Programs Director | Kansas Agricultural Experiment Station
Cell: 701-630-0464 Email: jmschuh@ksu.edu

Project Personnel and Partner Organizations

1. David C. Molik; Kansas State University; Principal Investigator
2. Nicholas Eggleston; Kansas State University; Co-Principal Investigator
3. Jeffrey Sheldon; Kansas State University; Co-Principal Investigator
4. Gwen Sibley; Kansas State University; Senior Personnel
5. Taylor Chestnut; Kansas State University, Office of Educational Innovation and Evaluation; Evaluation Lead and K-FORGE Advisory Committee Member
6. Adrian Self; Kansas State University, National Agricultural Biosecurity Center; K-FORGE Advisory Committee Member and Unpaid Collaborator
7. Daniel Andresen; Kansas State University, Department of Computer Science and Institute for Computational Research in Engineering and Science; K-FORGE Advisory Committee Member and Unpaid Collaborator
8. Susan Metzger; Kansas State University, Office of the Vice President for Research, Institute for Digital Agriculture and Advanced Analytics, and Kansas Water Institute; K-FORGE Advisory Committee Member and Unpaid Collaborator
9. Mickey Slimp; Great Plains Network; K-FORGE Advisory Committee Member and Unpaid Collaborator
10. Jane M. Schuh; Kansas State University, Kansas Agricultural Experiment Station and College of Agriculture; K-FORGE Advisory Committee Member and Unpaid Collaborator
11. LeAnn Brosius; Kansas State University, Office of Educational Innovation and Evaluation; Evaluation Partner Institutional Representative and Writer of Letter of Collaboration
12. Marty Vanier; Kansas State University, National Agricultural Biosecurity Center; Security Partner Institutional Representative and Writer of Letter of Collaboration

DETAILED COST ESTIMATE

Project Period: February 1, 2027-January 31, 2030 | Total Request: $9,000,000

1. Purpose and Estimating Framework

This supplementary document provides the detailed cost estimate required for a Category II proposal under NSF 26-509. It presents all planned project costs across the three-year award, organized by project phase and by the organizational elements that will deliver the K-FORGE service. The estimate reconciles to the Research.gov budget: $6,422,965 in direct costs and $2,577,035 in facilities and administrative costs, for a total request of $9,000,000.

K-FORGE will integrate registry and repository services, managed workflow execution, research-computing infrastructure, instructional and workforce-development activities, evaluation, security review, and service-readiness planning into an operational national service. Year 1 emphasizes acquisition, development, integration, deployment, testing, and transition to full operations. Years 2 and 3 support continuous operations, national user onboarding, service improvement, evaluation, transferability assessment, and sustainability planning.

1.1 Estimating conventions and assumptions

- Salaries are escalated at 3 percent annually. Graduate tuition and selected operating costs are escalated at 5.5 percent annually.
- Equipment estimates are planning estimates for the functions and capacities described below. Final manufacturers, models, capacities, and component counts may change through technical design and competitive procurement while preserving the approved category totals and service capabilities.
- Equipment acquisition prices include warranty, replacement coverage, firmware and software support, and vendor maintenance necessary through the full project period ending January 31, 2030. No separate recurring equipment-maintenance charge is requested elsewhere in the budget.
- Kansas State University will provide routine electrical power, cooling, physical space, and ordinary data-center operations for K-FORGE equipment through existing institutional research-computing facilities. No direct costs for routine electricity, cooling, or space are requested from NSF, and these institutional resources are not assigned a monetary value in the proposal.
- The project will prioritize open-source software. No separate software-license line is requested. Any required cloud platform, storage, monitoring, backup, or security-service charges are included under Computer Services.
- No subawards, foreign travel, fee, or voluntary committed cost sharing are included. Dollar totals may reflect rounding to the nearest dollar.

2. Project Phases and Cost Summary

Project phase	Period	Direct costs	F&A	Total
Phase 1: Development, acquisition, deployment, and early operations	Year 1	$3,367,097	$997,475	$4,364,572
Phase 2: Continuous operations, national onboarding, and improvement	Year 2	$1,496,846	$773,799	$2,270,645
Phase 3: Continuous operations, optimization, transferability, and sustainability	Year 3	$1,559,022	$805,761	$2,364,783
Total	**Years 1-3**	**$6,422,965**	**$2,577,035**	**$9,000,000**

2.1 Organizational elements

Element 1 - Project leadership, coordination, and governance. Overall leadership, project coordination, administrative support, milestone and risk management, sponsor communication, advisory coordination, and project-wide reporting.

Element 2 - Registry, research-object, repository, and scholarly-record integration. Registry and API development; persistent identifiers; metadata, licensing, citation, contributor attribution, and versioning; research-object packaging; repository ingest; and durable linkage among workflows, run evidence, data, and publications.

Element 3 - Managed workflow execution and cyberinfrastructure operations. Beocat integration, workflow staging and execution, container and environment management, scheduling, storage, monitoring, logging, reliability, continuity, networking, heterogeneous accelerator verification, and production operations.

Element 4 - Training, workforce development, user support, and outreach. Graduate training, user onboarding, pilot support, workshops, demonstrations, online modules, and integration of reproducible computational practice into existing courses.

Element 5 - Evaluation, security, service readiness, dissemination, and sustainability. Independent evaluation, security and risk review, service-readiness and high-availability assessment, transferability and sustainability analysis, public reporting, documentation, and dissemination.

3. Labor Cost Basis

Labor estimates identify the number and type of personnel needed for development, continuous operations, user support, training, outreach, evaluation, security, and project administration. Senior/key personnel are budgeted at one calendar month per year each. The five professional positions and the administrative assistant are planned as full-time project appointments. The two graduate research assistants are budgeted at the institutionally approved graduate-assistant appointment level; their tuition is shown separately.

Personnel group, staffing basis, and element(s)	Year 1	Year 2	Year 3	Total
David Molik, PI				
Staffing basis: 1 person; 1 month/year				
Element 1	$9,013	$9,283	$9,561	$27,857
Nicholas Eggleston, Co-PI				
Staffing basis: 1 person; 1 month/year				
Element 3	$10,729	$11,051	$11,383	$33,163
Jeffrey Sheldon, Co-PI				
Staffing basis: 1 person; 1 month/year				
Element 4	$9,013	$9,283	$9,561	$27,857
Gwendolyn Sibley, Senior/Key Personnel				
Staffing basis: 1 person; 1 month/year				
Element 2	$6,867	$7,073	$7,285	$21,225
Sunderland instructor and project coordinator				
Staffing basis: 2 positions; 2.0 FTE/year				
Elements 1 and 4	$131,500	$135,445	$139,508	$406,453
Two research software engineers and one DevOps professional				
Staffing basis: 3 positions; 3.0 FTE/year				
Elements 2 and 3	$304,400	$313,532	$322,938	$940,870
Graduate research assistants				
Staffing basis: 2 appointments annually				
Elements 2, 3, and 4	$70,500	$72,615	$74,793	$217,908
Administrative assistant				
Staffing basis: 1 position; 1.0 FTE/year				
Element 1	$49,000	$50,470	$51,984	$151,454
Total salaries and wages	**$591,022**	**$608,752**	**$627,013**	**$1,826,787**
Fringe benefits				
Institutional rates by appointment type; all elements	$187,307	$192,927	$198,714	$578,948
Total compensation	**$778,329**	**$801,679**	**$825,727**	**$2,405,735**

3.1 Personnel roles and salary assumptions

The two 1.0-FTE positions supporting Sunderland and project coordination have a combined Year 1 salary estimate of $131,500, equivalent to an average planning salary of $65,750 per position before fringe benefits. Final salaries will be set through Kansas State University classification and hiring processes. The two research software engineers and one DevOps/cyberinfrastructure professional have a combined Year 1 salary estimate of $304,400, equivalent to an average planning salary of approximately $101,467 per position before fringe benefits. These positions require specialized software, systems, reliability, and research-cyberinfrastructure expertise.

The two graduate research assistants will form the initial cohort of K-State Libraries' Digital Scholarship Training Program. Their assignments will include workflow records, pilot documentation, validation, repository integration, training resources, benchmarking, and stakeholder feedback. Stipends total $70,500 in Year 1, or $35,250 per student, with 3 percent annual escalation. Tuition is budgeted separately at institutional rates.

4. Direct-Cost Schedule by Organizational Element and Phase

The following schedule assigns each direct-cost line to the organizational element or elements it principally supports and shows the phase in which the cost is incurred. Shared staff and project-wide costs are identified as cross-cutting rather than divided using unsupported allocation assumptions.

Cost item and element(s)	Year 1	Year 2	Year 3	Total
Senior/key personnel salaries (Elements 1-5)	$35,622	$36,690	$37,790	$110,102
Professional personnel: instructor, coordinator, engineers, DevOps (Elements 1-4)	$435,900	$448,977	$462,446	$1,347,323
Graduate research assistant stipends (Elements 2-4)	$70,500	$72,615	$74,793	$217,908
Administrative assistant salary (Element 1)	$49,000	$50,470	$51,984	$151,454
Fringe benefits (Elements 1-5)	$187,307	$192,927	$198,714	$578,948
Edge-AI reproducibility testbed (Element 3)	$400,000	$0	$0	$400,000
Repository and workflow storage upgrade (Elements 2-3)	$750,000	$0	$0	$750,000
OpenStack repository and application service nodes (Element 2)	$300,000	$0	$0	$300,000
Domestic travel (Elements 1-5)	$36,000	$37,980	$40,069	$114,049
Participant support (Element 4)	$14,000	$14,620	$15,269	$43,889
Materials and supplies (Elements 1-4)	$52,700	$0	$0	$52,700
Publication, documentation, and dissemination (Elements 1, 4-5)	$12,000	$12,660	$13,356	$38,016
Consultant and professional services (Element 5)	$140,000	$147,700	$155,824	$443,524
Cloud computer services and warm continuity environment (Element 3)	$98,000	$206,780	$218,202	$522,982
Graduate tuition (Elements 2-4)	$21,068	$22,227	$23,449	$66,744
OEIE evaluation (Element 5)	$200,000	$211,000	$222,605	$633,605
Second Beocat network-line upgrade (Element 3)	$25,000	$0	$0	$25,000
NABC security support (Element 5)	$40,000	$42,200	$44,521	$126,721
Data-center power redundancy and HA support (Element 3)	$500,000	$0	$0	$500,000
Total direct costs	**$3,367,097**	**$1,496,846**	**$1,559,022**	**$6,422,965**

5. Basis of Estimate for Major Cost Areas

5.1 Equipment acquisition - $1,450,000 in Year 1

Equipment will establish the operational service environment required to enter full operations by the end of Year 1. Acquisition estimates include primary hardware, required memory and storage, networking and optics, rack integration, remote management, spare capacity, and warranty or vendor support through January 31, 2030. Final purchases will be competitively procured and technically coordinated with Beocat, university information-technology staff, facilities personnel, and Sponsored Programs.

Low-power edge-AI reproducibility testbed - $400,000. The planning configuration is a 24-node, rack-integrated, low-power testbed, anticipated as approximately six four-node chassis or an equivalent serviceable configuration. Eight nodes will represent each of three accelerator classes - TPU, FPGA, and NPU - with approximately four M.2 accelerator positions per node, providing 96 total accelerator positions. The testbed supports controlled verification of workflow compatibility, portability, telemetry capture, and reproducibility across emerging accelerator environments; it is not a stand-alone general-purpose AI instrument.

Planning component	Quantity/basis	Estimate
Rack-integrated chassis and 24 compute nodes	Approximately six four-node chassis or equivalent	$192,000
Representative TPU-, FPGA-, and NPU-class accelerators	Approximately 96 M.2 devices	$96,000
Management/head node	One production-grade management system	$20,000
Switching, network adapters, and optics	Data and management connectivity	$32,000
Rack integration, cabling, power distribution, spares, and warranty	Installation and three-year support allowance	$60,000
Total		**$400,000**

Repository and workflow storage upgrade - $750,000. The storage upgrade will support active workflow staging, large reference datasets, software and container artifacts, outputs, logs, run-level provenance, benchmark artifacts, repository ingest, backup, and handoff to public-access systems. The architecture must support both large sequential transfers and high volumes of smaller metadata, log, and provenance objects. Final capacity will balance usable storage, throughput, redundancy, rebuild time, maintainability, and integration with existing Beocat and repository storage.

Planning component	Quantity/basis	Estimate
Storage servers and enclosures	Redundant production storage systems	$180,000
High-capacity storage media	Active workflow, artifact, output, and staging capacity	$350,000
NVMe/cache/metadata tier	High-I/O metadata, provenance, and small-object workloads	$80,000
High-throughput networking and optics	Storage, Beocat, and repository connectivity	$55,000
Replication, backup, spare media, and replacement components	Reliability and rebuild support	$35,000
Rack integration and vendor support	Installation and support through the project period	$50,000
Total		**$750,000**

OpenStack repository and application service nodes - $300,000. These nodes will provide a redundant production environment for registry, repository, API, identity, metadata, monitoring, database, and supporting services. Persistent control and application functions will be separated from transient research workloads on Beocat so that the public-facing service and execution environment can scale and be maintained on different operational schedules.

Planning component	Quantity/basis	Estimate
Application and compute nodes	Approximately six production nodes	$180,000
Control, database, identity, and monitoring nodes	Approximately three redundant service nodes	$75,000
Network interfaces, switching, and optics	Application and storage connectivity	$20,000
Spares, rack integration, and vendor support	Reliability and support through the project period	$25,000
Total		**$300,000**

5.2 Managed operations, connectivity, continuity, and power

Computer Services total $522,982: $98,000 in Year 1, $206,780 in Year 2, and $218,202 in Year 3. Year 1 supports architecture, configuration, and approximately a partial year of warm continuity operations. Years 2 and 3 support full-year operations, with 5.5 percent escalation. Services may include selected registry and application compute, replicated object or block storage, snapshots and backup, network traffic, monitoring, logging, security services, and limited failover or recovery exercises. The cloud environment will preserve selected critical functions and recovery data; it will not duplicate the full Beocat execution platform.

The Year 1 **Second Beocat Network-Line Upgrade** estimate of $25,000 covers project-attributable optics, configuration, activation, testing, and related work for a second network path. The basis is the availability of an existing fiber path that primarily requires optics and configuration rather than construction of a new external connection. Final scope will be coordinated with campus networking and KanREN.

The Year 1 **Data-Center Power Redundancy and High-Availability Support** estimate of $500,000 is a planning allowance based on preliminary data-center and facilities assessment of the project-specific work required to reduce a single point of electrical failure. Allowable scope may include engineering; redundant uninterruptible-power, transfer, distribution, or rack-level components; installation and commissioning; monitoring integration; and testing. Final scope and pricing will be established after the production equipment configuration and measured power requirements are confirmed. The estimate excludes routine electricity, routine cooling, general building renovation, unrelated deferred maintenance, and expansion of campus utility capacity beyond project-specific needs.

Routine power and cooling will be provided by Kansas State University through existing institutional operations and will not be charged to the award. The project will document typical and peak load after final equipment selection to confirm deployment within the facility operating envelope.

5.3 Training, outreach, participant support, and instructional integration

Participant Support totals $43,889: $14,000 in Year 1, $14,620 in Year 2, and $15,269 in Year 3. The estimate anticipates approximately 106 eligible non-employee participants annually, based on participation and attendance patterns from comparable prior advisory, workshop, demonstration, and community activities. Year 1 includes $6,000 for participant travel and $8,000 for other allowable participation expenses; these amounts increase to $6,180 and $8,440 in Year 2 and to $6,365 and $8,904 in Year 3. Support will combine a smaller number of travel awards with modest costs paid directly to or on behalf of a broader group of eligible participants. Project employees and individuals performing project work will not be charged to participant support.

Students and faculty reached through online modules or existing courses are broader-impact and education participants, but they will not be counted in the participant-support total unless an allowable cost is paid directly to or on their behalf. This distinction permits the instructional program to reach substantially more learners than the 106 annual participant-support recipients.

Materials and Supplies total $52,700 in Year 1: $24,000 for project-personnel workstations, $25,200 for the Sunderland instructional and workflow-prototyping laboratory and representative accelerator devices, and $3,500 for PI computing equipment. Each item is below the institutional equipment threshold.

Publication, Documentation, and Dissemination total $38,016 across the project and will support accessible online technical documentation, online modules and training materials, public implementation guidance, the public evaluation report, and appropriate open-access dissemination.

Domestic Travel totals $114,049: $36,000 in Year 1, $37,980 in Year 2, and $40,069 in Year 3. The Year 1 base covers participation in an annual NSF PI meeting or major cyberinfrastructure conference, technical and project-team coordination, pilot and community engagement, dissemination, and limited site or partner visits. Annual amounts are escalated at 5.5 percent. No foreign travel is requested.

5.4 Evaluation, security, service readiness, and sustainability

The **Office of Educational Innovation and Evaluation** estimate totals $633,605: $200,000 in Year 1, $211,000 in Year 2, and $222,605 in Year 3. The estimate supports an independent professional evaluation team to design the evaluation framework; establish baselines, indicators, and data responsibilities; administer surveys and structured feedback; synthesize advisory committee

findings; analyze service and collaboration evidence; provide formative recommendations; and prepare annual, summative, and public evaluation products. The Year 1 base is escalated at 5.5 percent.

National Agricultural Biosecurity Center security support totals $126,721: $40,000 in Year 1, $42,200 in Year 2, and $44,521 in Year 3. The estimate supports threat and risk assessment, research-security review, physical-access and asset-protection review, workflow-onboarding security requirements, incident and escalation procedures, review of sensitive-data pilots, and annual reassessment of the service risk posture. The Year 1 base is escalated at 5.5 percent.

Consultant and Professional Services total $443,524: $140,000 in Year 1, $147,700 in Year 2, and $155,824 in Year 3. This annual bounded allowance supports competitively procured, specialized external services not available within the project team, including service-readiness validation, high-availability architecture review, recovery and continuity assessment, operational and sustainability analysis, transferability review, and specialized engineering or security consultation. Work will be acquired through fixed-price or otherwise bounded statements of work with defined deliverables. No consultant will direct the project or duplicate work assigned to university employees, NABC, OEIE, or Beocat.

5.5 Graduate tuition and other direct-cost assumptions

Graduate tuition totals $66,744: $21,068 in Year 1, $22,227 in Year 2, and $23,449 in Year 3 for the two graduate research assistants. The Year 1 estimate follows institutional tuition assumptions and is escalated at 5.5 percent annually. Tuition is excluded from the modified total direct cost base in accordance with the university's negotiated rate agreement.

No subawards are included. Project partners participate through Kansas State University units, advisory roles, or unfunded collaboration. The project does not request funds for individual research enabled by the infrastructure.

6. Total Cost of Ownership and Sustainability Factors

The principal award-period Total Cost of Ownership factors are personnel for continuous operations and user support; storage and application infrastructure; cloud continuity services; warranty and maintenance coverage; networking; security governance; evaluation; and project-specific high-availability work. These costs are explicitly represented in the estimate. Routine institutional power, cooling, space, and ordinary data-center operations are available through Kansas State University and are not charged to NSF.

The project will track utilization, service performance, staffing effort, storage growth, cloud consumption, maintenance requirements, and support demand throughout the award. Those records, together with independent evaluation and external service-readiness review, will inform the operating model, transferability analysis, and sustainability plan. Equipment and services are scoped to establish a production service during the award while preserving procurement flexibility and avoiding unsupported assumptions about post-award expansion.

LIST OF SUGGESTED REVIEWERS

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List of Reviewers Not to Include

Data Not Available